

Unsupervised Pre-training for Detection Transformers

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Abstract—DEtection TRansformer (DETR) for object detection reaches competitive performance compared with Faster R-CNN via a transformer encoder-decoder architecture. However, trained with scratch transformers, DETR needs large-scale training data and an extreme long training schedule even on COCO dataset. Inspired by the great success of pre-training transformers in natural language processing, we propose a novel pretext task named random query patch detection in Unsupervised Pre-training DETR (UP-DETR). Specifically, we randomly crop patches from the given image and then feed them as queries to the decoder. The model is pre-trained to detect these query patches from the input image. During the pre-training, we address two critical issues: multi-task learning and multi-query localization. (1) To trade off classification and localization preferences in the pretext task, we find that freezing the CNN backbone is the prerequisite for the success of pre-training transformers. (2) To perform multi-query localization, we develop UP-DETR with multi-query patch detection with attention mask. Besides, UP-DETR also provides a unified perspective for fine-tuning object detection and one-shot detection tasks. In our experiments, UP-DETR significantly boosts the performance of DETR with faster convergence and higher average precision on object detection, one-shot detection and panoptic segmentation. Code and pre-training models: <https://github.com/ddzg/up-detr>.

Index Terms—Transformer, Unsupervised Pre-training, Self-supervised Learning, Object Detection, One-shot Detection

1 INTRODUCTION

DETECTION TRansformer (DETR) [1] is a recent framework that views object detection as a direct set prediction problem via a transformer encoder-decoder [2]. Without hand-designed sample selection [3] and non-maximum suppression (NMS), DETR even reaches a competitive performance with Faster R-CNN [4]. However, DETR comes with training and optimization challenges, which need a large-scale training dataset and an extremely long training schedule even on COCO dataset [5]. Besides, it is found that DETR performs poorly in PASCAL VOC dataset [6] which has insufficient training data and fewer instances than COCO. Fig. 1 shows the PASCAL VOC learning curves of DETR and our **Unsupervised Pre-training DETR (UP-DETR)**. UP-DETR converges much faster with a higher AP than DETR.

With well-designed pretext tasks, unsupervised pre-training models achieve remarkable progress in both natural language processing (e.g. GPT [7], [8] and BERT [9]) and computer vision (e.g. MoCo [10], [11] and SwAV [12]). In DETR, the CNN backbone (ResNet-50 [13] with $\sim 23.2M$ parameters) has been pre-trained to extract a good visual representation, but the transformer module with $\sim 18.0M$ parameters has not been pre-trained. Therefore, it is a straightforward idea to pre-train the transformer in DETR.

Although unsupervised visual representation learning (e.g. contrastive learning) attracts much attention in recent studies [10], [14], [15], [16], [17], [18], existing pretext tasks can not directly apply to pre-train the transformer in DETR. We summarize two main reasons from the following perspectives:

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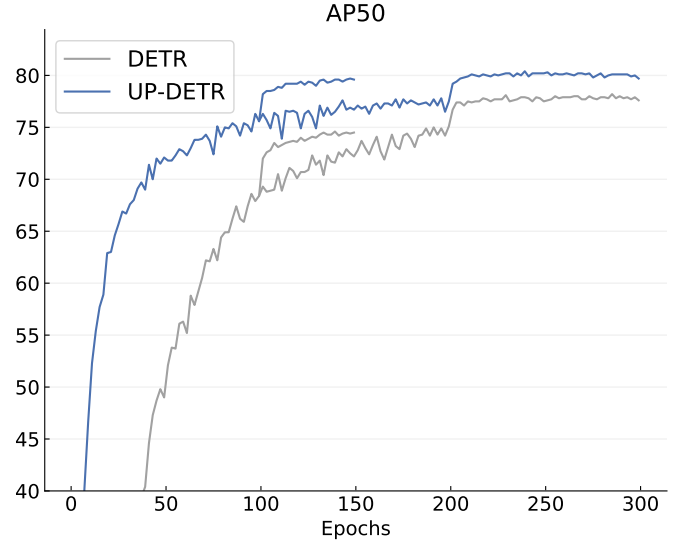


Fig. 1: The PASCAL VOC learning curves (AP_{50}) of DETR and UP-DETR with the ResNet-50 backbone. Here, they are trained on `trainval07+12` and evaluated on `test2007`. We plot the short and long training schedules, and the learning rate is reduced at 100 and 200 epochs respectively for these two schedules. UP-DETR converges much faster with a higher AP than DETR.

- (1) **Different architectures:** Recently, popular unsupervised learning methods are designed for the backbone (feature extractor) pre-training [10], [14], [15] to extract the image feature. However, DETR consists of a backbone and a transformer encoder-decoder. The transformer in DETR is designed to translate the feature into a set of detection targets instead of extracting the visual representation.

无监督预训练用于检测变换器

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摘要——基于Transformer编码器-解码器架构的目标检测模型DEtection TRansformer (DETR) 已达到与Faster R-CNN相媲美的性能。然而, 由于采用从头训练的Transformer结构, DETR即使在COCO数据集上也需要大规模训练数据和极长的训练周期。受Transformer在自然语言处理领域预训练取得巨大成功的启发, 我们在无监督预训练DETR (UP-DETR) 中提出了一种名为随机查询图像块检测的新颖预训练任务。具体而言, 我们从给定图像中随机裁剪图像块作为解码器的查询输入, 通过预训练使模型学会从原始图像中检测这些查询图像块。在预训练过程中, 我们解决了两个关键问题: 多任务学习与多查询定位。(1) 为权衡预训练任务中分类与定位的偏好, 我们发现冻结CNN骨干网络是成功预训练Transformer的前提条件。(2) 为实现多查询定位, 我们开发了基于注意力掩码的多查询图像块检测机制。此外, UP-DETR为目标检测和单样本检测任务提供了统一的微调框架。实验表明, UP-DETR显著提升了DETR在目标检测、单样本检测和全景分割任务上的性能, 实现了更快的收敛速度和更高的平均精度。代码与预训练模型: <https://github.com/dddzg/up-detr>。

索引术语—Transformer, 无监督预训练, 自监督学习, 目标检测, 单次检测

1 引言

DETR (Detection TRansformer) [1] 是近期提出的一种框架, 它将目标检测视为通过Transformer编码器-解码器[2]直接进行集合预测的问题。无需人工设计的样本选择[3]和非极大值抑制 (NMS), DETR甚至能达到与Faster R-CNN[4]相竞争的性能。然而, DETR存在训练和优化方面的挑战, 即使是在COCO数据集[5]上, 也需要大规模训练数据和极长的训练周期。此外, 研究发现DETR在训练数据不足且实例数量少于COCO的PASCAL VOC数据集[6]上表现不佳。图1展示了DETR与我们提出的无监督预训练DETR (UP-DETR) 在PASCAL VOC上的学习曲线。UP-DETR以更高的平均精度 (AP) 实现了比DETR快得多的收敛速度。

通过精心设计的预训练任务, 无监督预训练模型在自然语言处理 (e.g. GPT [7]、[8] 和 BERT [9]) 与计算机视觉 (e.g. MoCo [10]、[11] 和 SwAV [12]) 领域均取得了显著进展。在DETR中, CNN骨干网络 (ResNet-50 [13], 参数量为 ~ 23.2 M) 已通过预训练获得良好的视觉特征提取能力, 但参数量为 ~ 18.0 M的Transformer模块尚未经过预训练。因此, 对DETR中的Transformer进行预训练是一个直接的思路。

尽管无监督视觉表示学习 (e.g. 对比学习) 在近期研究中备受关注[10], [14], [15], [16], [17], [18], 但现有的预训练任务无法直接应用于DETR中transformer的预训练。我们从以下角度总结出两个主要原因:

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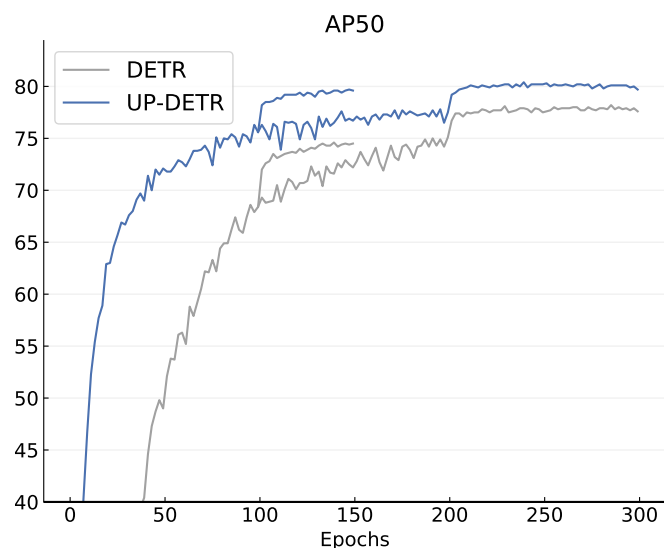


图1: 采用ResNet-50骨干网络的DETR与UP-DETR在PASCAL VOC数据集上的学习曲线 (AP_{50})。模型在trainval07+12上训练, 并在test2007上评估。我们绘制了短期与长期训练计划的学习曲线, 两种计划的学习率分别在100和200轮次时下降。UP-DETR以更高的AP值实现了比DETR快得多的收敛速度。

(1) **Different architectures:** 近期流行的无监督学习方法主要针对骨干网络 (特征提取器) 的预训练设计[10], [14], [15], 以提取图像特征。然而, DETR由骨干网络和Transformer编码器-解码器构成。其中的Transformer旨在将特征转化为一组检测目标, 而非提取视觉表征。

- (2) **Different feature preference:** The transformer in DETR mainly focuses on spatial localization learning. Specifically, the transformer is used to translate the instance-level information of the image into coordinates. As for the self-attention in the decoder, it performs a NMS-like mechanism to suppress duplicated bounding boxes. However, existing pretext tasks are designed for image instance-based [10], [14], [15] or cluster-based [16], [17], [18] contrastive learning. These contrastive learning methods mainly focus on feature discrimination rather than spatial localization.

As analyzed above, we need to construct a spatial localization related task to pre-train the transformer in DETR. To realize this idea, we propose an Unsupervised Pre-training DETR (UP-DETR) with a novel unsupervised pretext task named **random query patch detection** to pre-train the detector without any human annotations — we *randomly* crop multiple *query patches* from the given image, and pre-train the transformer for object *detection* to predict the bounding boxes of these query patches in the given image. During the pre-training procedure, we address two critical issues as follows:

- (1) **Multi-task learning:** Object detection is the coupling of object classification and localization. To avoid query patch detection destroying the classification features, we introduce the **frozen pre-training backbone** and patch feature reconstruction to preserve the feature discrimination of the transformer. In our experiments, we find that the frozen pre-training backbone is the most important step to preserve the feature discrimination during the pre-training.
- (2) **Multi-query localization:** Different object queries focus on different position areas and box sizes. There is a NMS-like mechanism between different object queries. Considering this property, we propose a **multi-query patch** detection task with attention mask.

Besides, the proposed UP-DETR also provides a unified perspective for object detection and one-shot detection. Just changing the input of the decoder, we can easily fine-tune UP-DETR on object detection and one-shot detection. In our experiments, we pre-train DETR with the random query patch detection task. Our UP-DETR performs much better than DETR using exactly the same architecture on PASCAL VOC [6] and COCO [5] object detection datasets with faster convergence speed and better average precision. Furthermore, UP-DETR also transfers well with state-of-the-art performance on one-shot detection and panoptic segmentation.

2 RELATED WORK

2.1 Object Detection

Positive and negative sample assignment is an important component for object detection frameworks. Two-stage detectors [4], [19] and a part of one-stage detectors [20], [21] construct positive and negative samples by hand-crafted multi-scale anchors with the IoU threshold and model confidence. Anchor-free one-stage detectors [22], [23], [24] assign positive and negative samples to feature maps by a grid of object centers. Zhang *et al.* demonstrate that the performance gap between them is due to the selection of positive and negative training samples [3]. DETR [1] is a recent object detection framework that is conceptually simpler without hand-crafted process by direct set prediction [25], which assigns the positive and negative samples automatically.

Apart from the positive and negative sample selection problem, the trade-off between classification and localization is also intractable for object detection. Zhang *et al.* illustrate that there is a domain misalignment between classification and localization [26]. Wu *et al.* [27] and Song *et al.* [28] design two head structures for classification and localization. They point out that these two tasks may have opposite feature preferences. As for our pre-training model, it shares feature for classification and localization. Therefore, it is essential to take a well trade-off between these two tasks.

2.2 Unsupervised Pre-training

Unsupervised pre-training models always follow two steps: pre-training on a large-scale dataset with the pretext task and fine-tuning the parameters on downstream tasks. For unsupervised pre-training, the pretext task is always invented, and we are interested in the learned intermediate representation rather than the final performance of the pretext task.

To perform unsupervised pre-training, there are various well-designed pretext tasks. For natural language processing, utilizing time sequence relationship between discrete tokens, masked language model [9], permutation language model [29] and auto regressive model [7], [8] are proposed to pre-train transformers [2] for language representation. For computer vision, unsupervised pre-training models also achieve remarkable progress recently for visual representation learning, outperforming the supervised learning counterpart in downstream tasks. Instance-based discrimination tasks [30], [31] and clustering-based tasks [17] are two typical pretext tasks in recent studies. Instance-based discrimination tasks vary mainly on maintaining different sizes of negative samples [10], [14], [15] with non-parametric contrastive learning [32]. Instance discrimination can also be realized as parametric instance classification [16]. Moreover, clustering-based tasks vary on offline [17], [18] or online clustering procedures [12]. Our proposed random query patch detection is a novel unsupervised pretext task, which is designed to pre-train the transformer based on the DETR architecture for object detection.

2.3 Vision Transformer

Transformers are not only used in detectors, but also applied to backbone designs. The pioneering work is Vision Transformer (ViT) [33], which applies a transformer on non-overlapping image patches for image classification. It achieves an speed-accuracy trade-off compared to convolutional neural networks with very large-scale training datasets. Based on this work, there are a lot of works that modify the ViT architecture for better image classification performance [34], [35], [36].

Though, ViT and DETR both use transformers, they are designed for different purposes. ViT uses transformers to capture the relation between image patches for image classification, so there is self-attention between image patches. On the other hand, there are three different attention parts in DETR. The self-attention in the encoder of DETR is similar to ViT, which is designed to capture global relation between image patches. However, the cross attention and self-attention in the decoder of DETR are designed for set prediction of boxes. Therefore, existing pre-training tasks of backbones [10], [11], [37] can not be simply applied to DETR.

(2) **Different feature preference**: DETR中的变压器主要专注于空间定位学习。具体来说, 变压器用于将图像的实例级信息转换为坐标。至于解码器中的自注意力机制, 它执行类似NMS的机制来抑制重复的边界框。然而, 现有的预训练任务是为基于图像实例的[10]、[14]、[15]或基于聚类的[16]、[17]、[18]对比学习设计的。这些对比学习方法主要关注特征区分而非空间定位。

如上所述, 我们需要构建一个与空间定位相关的任务来预训练DETR中的transformer。为实现这一构想, 我们提出了一种无监督预训练DETR (UP-DETR), 通过名为随机查询块检测的新型无监督前置任务, 在无需人工标注的情况下预训练检测器——我们从给定图像中*randomly*裁剪多个 *query patches*, 并预训练transformer以进行目标*detection*, 从而预测这些查询块在给定图像中的边界框。在预训练过程中, 我们重点解决了以下两个关键问题:

(1) **Multi-task learning**: 目标检测是目标分类与定位的结合。为避免查询补丁检测破坏分类特征, 我们引入冻结预训练主干网络和补丁特征重建机制, 以保持Transformer的特征判别力。实验发现, 冻结预训练主干网络是预训练过程中保持特征判别力最关键的一步。(2) **Multi-query localization**: 不同目标查询关注不同位置区域与框体尺寸。不同目标查询间存在类NMS机制。基于此特性, 我们提出采用注意力掩码的多查询补丁检测任务。

此外, 提出的UP-DETR还为物体检测和单样本检测提供了统一的视角。仅需改变解码器的输入, 我们就能轻松地在物体检测和单样本检测任务上对UP-DETR进行微调。在我们的实验中, 我们通过随机查询补丁检测任务对DETR进行预训练。在PASCAL VOC [6] 和 COCO [5] 物体检测数据集上, 我们的UP-DETR使用完全相同的架构, 相比DETR取得了显著更好的性能, 具有更快的收敛速度和更高的平均精度。此外, UP-DETR在单样本检测和全景分割任务上也表现出色, 达到了最先进的性能水平。

2 相关工作

2.1 目标检测

正负样本分配是目标检测框架中的一个重要组成部分。两阶段检测器[4]、[19]以及部分一阶段检测器[20]、[21]通过手工设计的多尺度锚框, 结合IoU阈值和模型置信度来构建正负样本。无锚框的一阶段检测器[22]、[23]、[24]则通过目标中心网格将正负样本分配到特征图上。Zhang *et al*的研究表明, 这些方法之间的性能差异主要源于正负训练样本的选择[3]。DETR[1]是近期提出的一种目标检测框架, 其概念更为简洁, 无需手工设计过程, 通过直接集合预测[25]自动完成正负样本的分配。

除了正负样本选择问题, 分类与定位之间的权衡对于目标检测来说也是一个棘手的问题。Zhang *et al* 指出分类与定位之间存在领域不对齐的问题[26]。Wu *et al* [27] 和 Song *et al* [28] 为分类和定位设计了两个头部结构。他们指出这两个任务可能具有相反的特征偏好。至于我们的预训练模型, 它共享分类和定位的特征。因此, 在这两个任务之间取得良好的权衡至关重要。

2.2 无监督预训练

无监督预训练模型通常遵循两个步骤: 首先在大规模数据集上通过前置任务进行预训练, 然后在下游任务上微调参数。对于无监督预训练而言, 前置任务往往是人为设计的, 我们关注的是学习到的中间表示, 而非前置任务的最终性能。

为了进行无监督预训练, 存在多种精心设计的代理任务。在自然语言处理领域, 利用离散标记间的时间序列关系, 研究者提出了掩码语言模型[9]、排列语言模型[29]以及自回归模型[7][8]来预训练用于语言表示的Transformer[2]。在计算机视觉领域, 无监督预训练模型近期在视觉表征学习方面也取得了显著进展, 在下游任务中表现超越了监督学习方法。基于实例的判别任务[30][31]和基于聚类的任务[17]是近期研究中的两类典型代理任务。基于实例的判别任务主要通过非参数对比学习[32]保持不同规模的负样本[10][14][15]来实现变体, 实例判别也可通过参数化实例分类[16]实现。此外, 基于聚类的任务可分为离线[17][18]与在线聚类流程[12]两类变体。我们提出的随机查询补丁检测是一种新型无监督代理任务, 其设计旨在基于DETR架构预训练用于目标检测的Transformer。

2.3 视觉变换器

Transformer不仅被用于检测器, 还应用于骨干网络的设计中。开创性的工作是Vision Transformer (ViT) [33], 它将Transformer应用于非重叠的图像块上进行图像分类。与需要超大规模训练数据集的卷积神经网络相比, ViT在速度与精度之间取得了平衡。基于这项工作, 后续出现了大量改进ViT架构以提升图像分类性能的研究[34]、[35]、[36]。

尽管ViT和DETR都使用了Transformer架构, 但它们的设计目标不同。ViT利用Transformer捕捉图像块之间的关系以进行图像分类, 因此图像块之间存在自注意力机制。而DETR则包含三种不同的注意力模块: 编码器中的自注意力与ViT类似, 旨在捕获图像块间的全局关联; 但解码器中的交叉注意力和自注意力是为边界框的集合预测而设计的。因此, 现有骨干网络的预训练任务[10]、[11]、[37]无法直接应用于DETR。

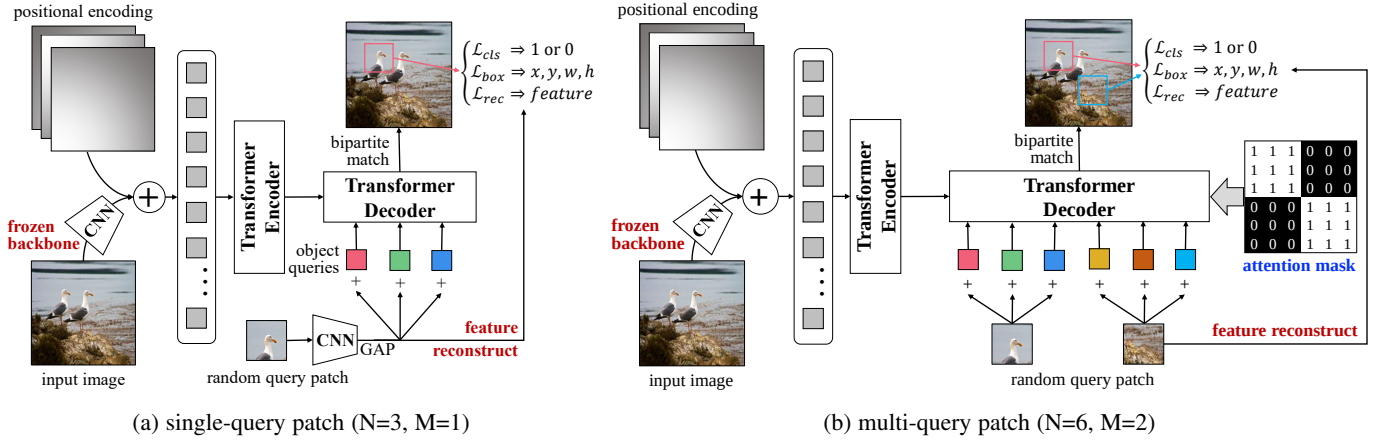


Fig. 2: The pre-training procedure of UP-DETR by random query patch detection. (a) There is only a single-query patch which we add to all object queries. (b) For multi-query patches, we add each query patch to N/M object queries with object query shuffle and attention mask. Note that CNN is not drawn in the decoder of (b) for neatness.

3 UP-DETR

The proposed UP-DETR contains pre-training and fine-tuning procedures: (a) the transformer is unsupervisedly *pre-trained* on a large-scale dataset (*e.g.* ImageNet in our experiments) without any human annotations; (b) the entire model is *fine-tuned* with labeled data which is similar as the original DETR [1] on the downstream detection related tasks (*e.g.* object detection and one-shot detection).

3.1 Pre-training

In this section, we mainly describe how to pre-train the transformer encoder and decoder with random query patch detection task. As shown in Fig. 2, the main idea of random query patch detection is simple but effective.

Encoder. Firstly, given an input image, a CNN backbone is used to extract the visual representation with the feature $f \in \mathbb{R}^{C \times H \times W}$, where C is the channel dimension and $H \times W$ is the feature map size. Here, an image can be treated as a token sequence with the length of $H \times W$ constructed by the CNN extractor. Then, the feature f is added with two-dimensional position encodings and passed to the multi-layer transformer encoder, which is exactly the same as DETR.

Decoder. Different from DETR, in our pre-training procedure, we randomly crop patches as the query from the input image and record the corresponding coordinates, width and height of the query as the ground truth. Therefore, our pre-training process can be done in an unsupervised/self-supervised paradigm. Specifically, for the random cropped query patch, the CNN backbone with global average pooling (GAP) extracts the patch feature $p \in \mathbb{R}^C$, which is flatten and supplemented with object queries $q \in \mathbb{R}^C$ before passing it into a transformer decoder. Finally, the decoder is trained to predict the bounding boxes corresponding to the position of random query patches in the input image. During the pre-training, the feature of query patch is added to multiple object queries, which are fed to the decoder. Note that the *query patch* refers to the cropped patch from the input image but *object query* refers to position embeddings. It can be understood as asking the model to find the query patch through these possible positions (embedded as object queries). Moreover, the CNN parameters of the input image and query patches are shared in the whole model.

Multi-query Patches. For fine-tuning object detection tasks, there are multiple object instances in each image (*e.g.* average 7.7 object instances per image in the COCO dataset). In other words, the self-attention module in the decoder transformer learns an NMS-like mechanism during the training. To make a consistency between pre-training and fine-tuning, UP-DETR needs to construct a competition between different object queries with an adjustable box suppression ratio. Fortunately, UP-DETR can be easily extended to multi-query patch detection. Assuming that there are M query patches by random cropping and N object queries, we divide N object queries into M groups, where each query patch is assigned to N/M object queries. In Fig. 2, we illustrate the single-query patch detection and multi-query patch detection. We adopt $N=100$ and $M=10$ by default in our pre-training experiments.

Matching. The model infers a prediction with a fixed-set $\hat{y} = \{\hat{y}_i\}_{i=1}^N$ corresponding to N object queries ($N > M$). In other words, we get N pairs of bounding box predictions for these M query patches. For detection tasks, the results are invariant to the permutation of predicted objects. Hence, following DETR [1], we compute the same match cost between the prediction $\hat{y}_{\hat{\sigma}(i)}$ and the ground-truth y_i using Hungarian algorithm [25], where $\hat{\sigma}(i)$ is the index of y_i computed by the optimal bipartite matching.

Loss. For the loss calculation, the predicted result $\hat{y}_i = (\hat{c}_i \in \mathbb{R}^2, \hat{b}_i \in \mathbb{R}^4, \hat{p}_i \in \mathbb{R}^C)$ consists of three elements: $\hat{c}_i$ is the binary classification of matching the query patch ($c_i = 1$) or not ($c_i = 0$) for each object query; $\hat{b}_i$ is the vector that defines the box center coordinates and its width and height as $\{x, y, w, h\}$, which are re-scaled relative to the image size; and $\hat{p}_i$ is the reconstructed feature with $C = 2048$ for the ResNet-50 backbone typically. With the above definitions, the Hungarian loss for all matched pairs is defined as:

$$\mathcal{L}(y, \hat{y}) = \sum_{i=1}^N [\lambda_{\{c_i\}} \mathcal{L}_{cls}(c_i, \hat{c}_{\hat{\sigma}(i)}) + \mathbb{1}_{\{c_i=1\}} \mathcal{L}_{box}(b_i, \hat{b}_{\hat{\sigma}(i)}) + \mathbb{1}_{\{c_i=1\}} \mathcal{L}_{rec}(p_i, \hat{p}_{\hat{\sigma}(i)})]. \quad (1)$$

Here, $\mathcal{L}_{cls}$ is the cross entropy loss over two classes (match the query patch *vs.* not match), and the class balance weight $\lambda_{\{c_i=1\}} = 1$ and $\lambda_{\{c_i=0\}} = M/N$. $\mathcal{L}_{box}$ is a linear combination of ℓ_1 loss and the generalized IoU loss with the same weight hyper-parameters as DETR [1]. $\mathcal{L}_{rec}$ is the reconstruction loss

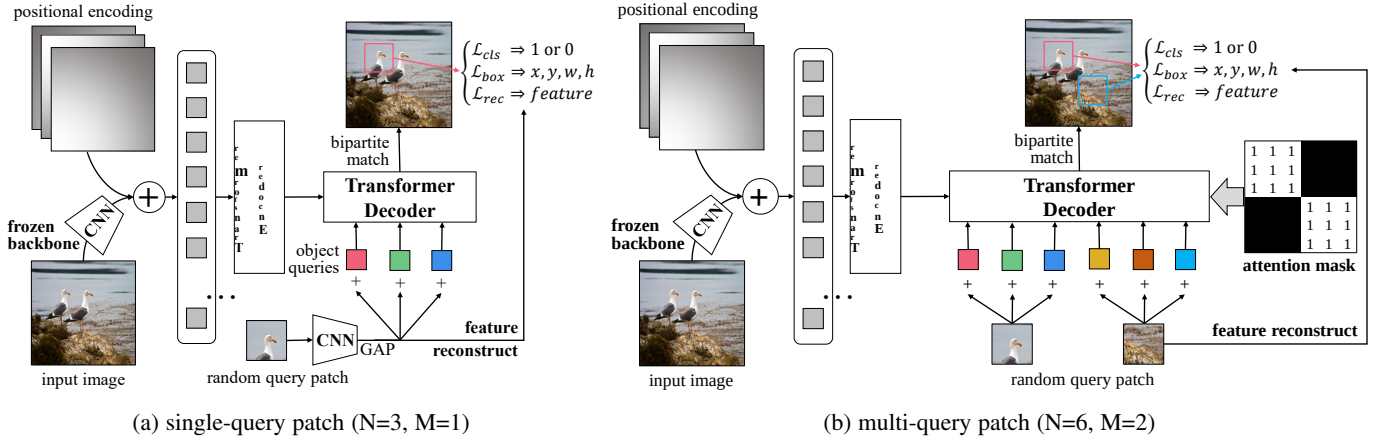


图2: 通过随机查询补丁检测进行UP-DETR预训练的过程。(a) 仅存在单个查询补丁, 我们将其添加到所有对象查询中。(b) 对于多查询补丁, 我们通过对象查询混洗和注意力掩码, 将每个查询补丁添加到 N/M 个对象查询中。注意: 为简洁起见, (b) 的解码器中未绘制CNN。

3 UP-DETR

提出的UP-DETR包含预训练和微调两个步骤: (a) 在大规模数据集 (e.g. 我们的实验中采用ImageNet) 上对transformer进行无监督*pre-trained*, 无需任何人工标注; (b) 在下游检测相关任务 (e.g. 目标检测与单样本检测) 上, 使用带标注数据对整个模型进行*fine-tuned*, 其方式与原始DETR[1]类似。

3.1 预训练

在本节中, 我们主要描述如何通过随机查询块检测任务预训练transformer编码器和解码器。如图2所示, 随机查询块检测的核心思想简单而有效。

编码器。首先, 给定输入图像, 使用CNN主干网络提取视觉表示, 其特征为 $f \in \mathbb{R}^{C \times H \times W}$, 其中 C 为通道维度, $H \times W$ 为特征图尺寸。此处, 图像可被视为由CNN提取器构建的长度为 $H \times W$ 的标记序列。随后, 特征 f 与二维位置编码相加, 并输入多层Transformer编码器, 其结构与DETR完全相同。

解码器。与DETR不同, 在我们的预训练过程中, 我们从输入图像中随机裁剪出补丁作为查询, 并记录查询对应的坐标、宽度和高度作为真实标签。因此, 我们的预训练过程可以在无监督/自监督范式下完成。具体来说, 对于随机裁剪的查询补丁, 采用全局平均池化 (GAP) 的CNN骨干网络提取补丁特征 $p \in \mathbb{R}^C$, 该特征在传入Transformer解码器前会被展平并与目标查询 $q \in \mathbb{R}^C$ 结合。最终, 解码器被训练用于预测输入图像中随机查询补丁对应位置的边界框。在预训练期间, 查询补丁的特征会添加到多个目标查询中, 并输入解码器。请注意, *query patch*指的是从输入图像中裁剪的补丁, 而*object query*指的是位置嵌入。这可以理解为要求模型通过这些可能的位置 (嵌入为目标查询) 来定位查询补丁。此外, 输入图像和查询补丁的CNN参数在整个模型中是共享的。

多查询补丁。对于微调目标检测任务, 每张图像中包含多个目标实例 (例如, COCO数据集中平均每张图像有7.7个目标实例)。换言之, 解码器Transformer中的自注意力模块在训练过程中学习了一种类似于非极大值抑制的机制。为了保持预训练与微调的一致性, UP-DETR需要构建不同目标查询之间的竞争关系, 并配备可调节的框抑制比例。幸运的是, UP-DETR可以轻松扩展至多查询补丁检测。假设通过随机裁剪生成 M 个查询补丁, 并设定 N 个目标查询, 我们将 N 个目标查询划分为 M 组, 其中每个查询补丁分配 N/M 个目标查询。图2展示了单查询补丁检测与多查询补丁检测的示意图。在预训练实验中, 我们默认采用 $N=100$ 和 $M=10$ 的配置。

匹配。模型通过一个固定集合 $\hat{y} = \{\hat{y}_i\}_{i=1}^N$ 推断出一个预测, 该集合对应于 N 个对象查询 ($N > M$)。换句话说, 我们为这些 M 个查询补丁得到 N 对边界框预测。对于检测任务, 结果对预测对象的排列顺序是不变的。因此, 遵循DETR [1] 的方法, 我们使用匈牙利算法 [25] 计算预测 $\hat{y}_{\hat{\sigma}(i)}$ 与真实标注 y_i 之间的相同匹配成本, 其中 $\hat{\sigma}(i)$ 是通过最优二分匹配计算出的 y_i 的索引。

损失。在损失计算中, 预测结果 $\hat{y}_i = (\hat{c}_i \in \mathbb{R}^2, \hat{b}_i \in \mathbb{R}^4, \hat{p}_i \in \mathbb{R}^C)$ 包含三个部分: $\hat{c}_i$ 是每个对象查询是否匹配查询图像块 ($c_i = 1$) 或不匹配 ($c_i = 0$) 的二分类结果; $\hat{b}_i$ 是定义边界框中心坐标及其宽度和高度的向量, 表示为 $\{x, y, w, h\}$, 这些值相对于图像尺寸进行了重新缩放; 而 $\hat{p}_i$ 是重建特征, 对于典型的ResNet-50骨干网络, 其维度 C 为2048。基于以上定义, 所有匹配对的匈牙利损失定义为:

$$\mathcal{L}(y, \hat{y}) = \sum_{i=1}^N [\lambda_{\{c_i\}} \mathcal{L}_{cls}(c_i, \hat{c}_{\hat{\sigma}(i)}) + \mathbb{1}_{\{c_i=1\}} \mathcal{L}_{box}(b_i, \hat{b}_{\hat{\sigma}(i)}) + \mathbb{1}_{\{c_i=1\}} \mathcal{L}_{rec}(p_i, \hat{p}_{\hat{\sigma}(i)})]. \quad (1)$$

这里, $\mathcal{L}_{cls}$ 是两类 (匹配查询图像块 *vs.* 与不匹配) 的交叉熵损失, 类别平衡权重分别为 $\lambda_{\{c_i=1\}}$ 和 $\lambda_{\{c_i=0\}} = M/N$ 。 $\mathcal{L}_{box}$ 是 ℓ_1 损失与广义IoU损失的线性组合, 其权重超参数与DETR [1] 保持一致。 $\mathcal{L}_{rec}$ 是重建损失。

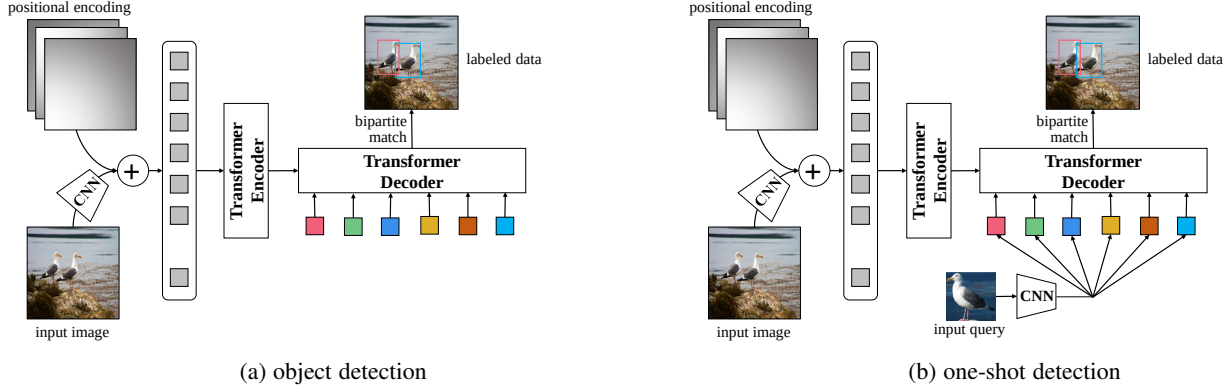


Fig. 3: The fine-tuning procedure of UP-DETR on object detection and one-shot detection. The only difference is the input of the transformer decoder. (a) Object detection. There are multiple object queries (learnable embeddings) fed into the decoder. (b) One-shot detection. The input query is extracted by the shared CNN, and the feature is added to object queries, which are fed into the decoder.

proposed in this paper and only used during the pre-training. It is designed to balance the feature preference of classification and localization during the pre-training, which will be further discussed in Section 3.1.2.

3.1.1 Frozen Pre-training Backbone

In our experiments, we find that the CNN backbone seriously affects the transformer pre-training. In other words, the model can not converge well and performs bad if we pre-train the CNN backbone and the transformer from scratch together with random query patch detection. This problem also appears in the original DETR¹.

Furthermore, object detection is the coupling of object classification and localization, where these two tasks always have different feature preferences [26], [27], [28]. However, our pre-training task only focuses on localization instead of feature classification. So, we freeze the pre-training CNN backbone during the transformer pre-training in our experiments. Stable backbone parameters are beneficial to the transformer pre-training, and accelerate the model pre-training process. In Section 4.5.2, we will analyze and verify the necessity of them with experiments. We argue that the frozen pre-training backbone is the most essential step to the success of UP-DETR.

3.1.2 Tricks

In our preliminary experiments, we tried some tricks with our prior knowledge in terms of random query patch detection. Here, we discuss two tricks during the pre-training, which are reasonable but not critical to the success of UP-DETR.

Patch Feature Reconstruction. Our pre-training task only focuses on localization instead of feature classification. In other words, missing an explicit branch for the classification task. So, with the frozen pre-training backbone, we propose a feature reconstruction loss term $\mathcal{L}_{rec}$ to preserve classification feature during localization pre-training. The motivation of this loss term is to preserve the feature discrimination extract by CNN after passing feature to the transformer. $\mathcal{L}_{rec}$ is the mean squared error between

the ℓ_2 -normalized patch feature extracted by the CNN backbone, which is defined as follows:

$$\mathcal{L}_{rec}(p_i, \hat{p}_{\hat{\sigma}(i)}) = \left\| \frac{p_i}{\|p_i\|_2} - \frac{\hat{p}_{\hat{\sigma}(i)}}{\|\hat{p}_{\hat{\sigma}(i)}\|_2} \right\|_2^2. \quad (2)$$

With the frozen CNN backbone, patch feature reconstruction slightly improves the fine-tuning performance as shown in Section 4.5.2.

Attention Mask. All the query patches are randomly cropped from the image. Therefore, they are independent without any relations. For example, the bounding box regression of the first cropping is not concerned with the second cropping. To satisfy the independence of query patches, we utilize an attention mask matrix to control the interactions between different object queries. The mask matrix $\mathbf{X} \in \mathbb{R}^{N \times N}$ is added to the softmax layer of self-attention [2] in the decoder as $\text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k} + \mathbf{X})\mathbf{V}$, where $\mathbf{Q} = \mathbf{K} = \mathbf{V}$ and they refer to the same set of the object query representation in the decoder. Similar to the token mask in UniLM [38], the attention mask is defined as:

$$X_{i,j} = \begin{cases} 0, & i, j \text{ in the same group} \\ -\infty, & \text{otherwise} \end{cases}, \quad (3)$$

where $X_{i,j}$ determines whether the object query q_i participates the interaction with the object query q_j . For intuitive understanding, the attention mask in Fig. 2b displays 1 and 0 corresponding to 0 and $-\infty$ in (3), respectively. Attention mask also slightly leads to a lower loss as demonstrated in Section 4.5.3.

3.2 Fine-tuning

UP-DETR provides a unified perspective in terms of detection related tasks. It can be easily fine-tuned into object detection or one-shot detection by changing the model input. Fig. 3 shows the fine-tuning procedure on object detection and one-shot detection. The CNN, transformer encoder-decoder and object queries are exactly the same for object detection and one-shot detection tasks, which are initialized from the pre-training procedure. The whole model weights are fine-tuned with the labeled data. The only difference in architecture between these two down-stream tasks is the input of transformer decoder. Noting that feature reconstruction loss is never used in the fine-tuning procedure.

Object detection. Given an input image, the model is required to predict to the set of objects with the bounding boxes and

1. <https://github.com/facebookresearch/detr/issues/157>

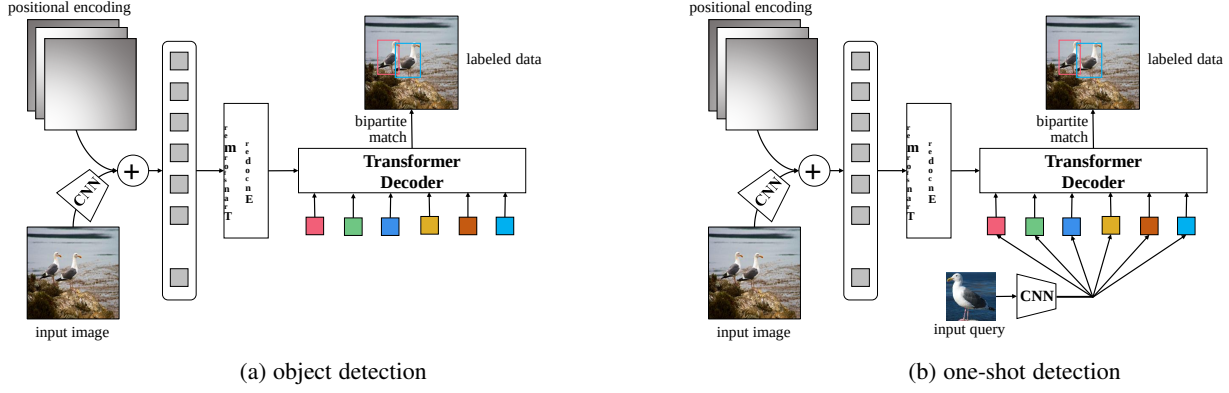


图3: UP-DETR在目标检测和单样本检测上的微调流程。唯一区别在于Transformer解码器的输入。(a) 目标检测。有多个目标查询(可学习嵌入向量)输入解码器。(b) 单样本检测。输入查询通过共享CNN提取, 其特征与目标查询相加后输入解码器。

本文提出并仅在预训练期间使用。它旨在平衡预训练期间分类和定位的特征偏好, 这将在第3.1.2节进一步讨论。

3.1.1 Frozen Pre-training Backbone

在我们的实验中, 我们发现CNN主干网络严重影响了transformer的预训练。换句话说, 如果我们使用随机查询补丁检测从头开始同时预训练CNN主干网络和transformer, 模型将无法良好收敛且表现不佳。这个问题在原始DETR¹中也同样存在。

此外, 目标检测是目标分类与定位的结合, 这两项任务始终具有不同的特征偏好[26]、[27]、[28]。然而, 我们的预训练任务仅侧重于定位而非特征分类。因此, 在实验中我们对预训练的CNN主干网络进行了冻结处理。稳定的主干网络参数有利于Transformer的预训练, 并加速模型预训练过程。在第4.5.2节中, 我们将通过实验分析和验证其必要性。我们认为, 冻结预训练主干网络是UP-DETR取得成功的最关键步骤。

3.1.2 Tricks

在我们初步的实验中, 我们利用先验知识在随机查询补丁检测方面尝试了一些技巧。这里, 我们将讨论预训练过程中的两个技巧, 这些技巧是合理的, 但对UP-DETR的成功并非关键。

补丁特征重建。我们的预训练任务仅关注定位而非特征分类。换言之, 缺失了用于分类任务的显式分支。因此, 在冻结预训练骨干网络的基础上, 我们提出了一个特征重建损失项 $\mathcal{L}_{rec}$, 以在定位预训练期间保留分类特征。该损失项的动机是在将特征传递给Transformer后, 保留CNN提取的特征判别力。 $\mathcal{L}_{rec}$ 是两者之间的均方误差。

由CNN骨干网络提取的 ℓ_2 归一化补丁特征, 其定义如下:

$$\mathcal{L}_{rec}(p_i, \hat{p}_{\hat{\sigma}(i)}) = \left\| \frac{p_i}{\|p_i\|_2} - \frac{\hat{p}_{\hat{\sigma}(i)}}{\|\hat{p}_{\hat{\sigma}(i)}\|_2} \right\|_2^2. \quad (2)$$

在冻结的CNN骨干网络下, 如第4.5.2节所示, 补丁特征重建略微提升了微调性能。

注意力掩码。所有查询补丁均从图像中随机裁剪获得, 因此它们相互独立且不存在任何关联。例如, 第一处裁剪的边界框回归与第二处裁剪无关。为满足查询补丁的独立性要求, 我们采用注意力掩码矩阵来控制不同对象查询间的交互作用。该掩码矩阵 $\mathbf{X} \in \mathbb{R}^{N \times N}$ 被添加至解码器自注意力机制[2]的softmax层中, 如 $\text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k} + \mathbf{X})\mathbf{V}$ 所示, 其中 $\mathbf{Q} = \mathbf{K} = \mathbf{V}$ 指向解码器中同一组对象查询表示。类似UniLM[38]中的令牌掩码机制, 注意力掩码定义为:

$$X_{i,j} = \begin{cases} 0, & i, j \text{ in the same group} \\ -\infty, & \text{otherwise} \end{cases}, \quad (3)$$

其中 $X_{i,j}$ 决定了对象查询 q_i 是否参与和对象查询 q_j 的交互。为便于直观理解, 图2b中的注意力掩码显示1和0, 分别对应公式(3)中的0和 $-\infty$ 。如第4.5.3节所示, 注意力掩码也略微降低了损失。

3.2 微调

UP-DETR为检测相关任务提供了一个统一的视角。通过改变模型输入, 它可以轻松微调用于目标检测或单样本检测。图3展示了在目标检测和单样本检测上的微调流程。CNN、Transformer编码器-解码器以及目标查询在目标检测和单样本检测任务中完全一致, 这些组件均从预训练过程中初始化。整个模型权重通过标注数据进行微调。这两个下游任务在架构上的唯一区别在于Transformer解码器的输入。需注意, 特征重建损失在微调过程中从不使用。

目标检测。给定输入图像, 模型需要预测带有边界框和

1. <https://github.com/facebookresearch/detr/issues/157>

the corresponding categories. Therefore, the fine-tuning procedure of UP-DETR is exactly the same with the training procedure of DETR. There are multiple object queries (learnable embeddings) used as the input fed into the decoder. Here, different object queries learn different spatial specialization and area preference. Benefiting from the design of object query, the model can predict multiple objects in parallel. With supervised object detection data, we compute the match cost between model prediction and the annotated ground-truth.

One-shot detection. Given an input image and a query image, the model is required to predict the objects with bounding boxes. The objects should be semantically similar to the query image. Typically, the query image is constructed by patch from different image with the same category. For one-shot detection, the query image is extracted by the CNN (shared with the input image), and the patch feature is added to all object queries. In one-shot detection, we only care about the matching result (*i.e.* match to the query or not) of the bounding boxes instead of the specific category in the object detection. So, in the loss and match cost computation, there is a binary classification output in one-shot detection instead of multiple category classification in object detection.

4 EXPERIMENTS

We pre-train the model using ImageNet [39] and fine-tune the parameters on PASCAL VOC [6] and COCO [5]. In all experiments, we implement the UP-DETR model (41.3M parameters) with ResNet-50 backbone, 6 transformer encoder, 6 decoder layers of width 256 with 8 attention heads. Referring to the open source code of DETR², we use the same hyper-parameters in the proposed UP-DETR and our DETR re-implementation. We annotate R50 and R101 short for ResNet-50 and ResNet-101. Note that UP-DETR and DETR have exactly the same model architecture, match cost and loss calculation. So, they have exactly the same FLOPs, parameters and running FPS.

Pre-training setup. UP-DETR is unsupervisedly pre-trained on the 1.28M ImageNet training set without any ImageNet labels. The CNN backbone (ResNet-50) is also unsupervisedly pre-trained with SwAV strategy [12], and its parameters are frozen during the transformer pre-training. As the input image from ImageNet is relatively small, we resize it such that the shortest side is within [320, 480] pixels while the longest side is at most 600 pixels. For the given image, we crop the query patches with random coordinate, height and width, which are resized to 128×128 pixels and transformed with SimCLR style [14] without horizontal flipping, including random color distortion and Gaussian blurring. In the early stage of our work, we conducted experiments about different sampling strategies, *i.e.*, grid sampling, and random sampling with different hyper-parameters. However, we noticed that there were no significant differences between different sampling strategies. Therefore, we use random sampling to illustrate the generalization capability of our method. Moreover, AdamW [40] is used to optimize the UP-DETR, with the initial learning rate of 1×10^{-4} and the weight decay of 1×10^{-4} . We use a mini-batch size of 256 on eight Nvidia V100 GPUs to train the model for 60 epochs with the learning rate multiplied by 0.1 at 40 epochs.

Fine-tuning setup. The model is initialized with the pre-trained UP-DETR parameters and fine-tuned for all the parameters (including CNN) on PASCAL VOC and COCO. We fine-tune the

TABLE 1: Object detection results trained on PASCAL VOC trainval07+12 and evaluated on test2007. DETR and UP-DETR use R50 backbone and Faster R-CNN uses R50-C4 backbone. The values in the brackets are the gaps compared to DETR with the same training schedule.

Model/Epoch	AP	AP ₅₀	AP ₇₅
Faster R-CNN	56.1	82.6	62.7
DETR/150	49.9	74.5	53.1
UP-DETR/150	56.1 (+6.2)	79.7 (+5.2)	60.6 (+7.5)
DETR/300	54.1	78.0	58.3
UP-DETR/300	57.2 (+3.1)	80.1 (+2.1)	62.0 (+3.7)

model with the initial learning rate of 1×10^{-4} for the transformer and 5×10^{-5} for the CNN backbone, and the other settings are the same as DETR [1] on eight V100 GPUs with four images per GPU. The model is fine-tuned with short/long schedule for 150/300 epochs and the learning rate is multiplied by 0.1 at 100/200 epochs, respectively.

Gap between pre-training and fine-tuning. Object-level pre-training can boost the fine-tuning performance in object detection task [41], but such method usually requires object prior knowledge, *e.g.*, using a selective search algorithm to get object proposals during pre-training. In our work, we pre-train the model on ImageNet dataset at patch level, as this is a common practice in pre-training on ImageNet and transferring to COCO or VOC datasets [10], [37]. However, the gap between pre-training and fine-tuning always exists. For example, ImageNet images mostly contain one object, but COCO images always contain multiple objects. The image size of ImageNet images is relatively small, but the image size of COCO images is large. Moreover, the pre-training task is conducted at patch level with random sampling in this work, but the fine-tuning task is supervised on labeled data. Although the aforementioned gap between pre-training and fine-tuning cannot be eliminated, our method has good generalization capability using random sampling strategy without any prior knowledge.

4.1 PASCAL VOC Object Detection

Setup. The model is fine-tuned on PASCAL VOC trainval07+12 ($\sim 16.5k$ images) and evaluated on test2007. We report COCO-style metrics, including AP, AP₅₀ (default VOC metric) and AP₇₅. For a full comparison, we also report the result of Faster R-CNN with the R50-C4 backbone [12], which performs much better than R50 (C5 stage) [42]. DETR with R50-C4 significantly increases the computational cost as compared with R50, so we fine-tune UP-DETR with R50 backbone. Even though, UP-DETR still performs well. To emphasize the effectiveness of pre-training models, we report the results of 150 and 300 epochs for both DETR (from the random initialized transformer) and UP-DETR (from the pre-trained transformer).

Results. Table 1 shows the object detection results on PASCAL VOC dataset. We find that DETR performs poorly on PASCAL VOC, which is much worse than Faster R-CNN by a large gap in all metrics. Due to the relatively small-scale data in VOC, the pre-training transformer of UP-DETR significantly boosts the performance of DETR for both short and long schedules: up to **+6.2** (+3.1) AP, **+5.2** (+2.1) AP₅₀ and **+7.5** (+3.7) AP₇₅ for 150 (300) epochs, respectively. Moreover, UP-DETR (R50) achieves a comparable result to Faster R-CNN (R50-C4) with better AP. We

2. <https://github.com/facebookresearch/detr>

对应的类别。因此，UP-DETR的微调过程与DETR的训练过程完全相同。有多个对象查询（可学习的嵌入）作为输入馈送到解码器中。在这里，不同的对象查询学习不同的空间专业化和区域偏好。得益于对象查询的设计，模型可以并行预测多个对象。利用有监督的目标检测数据，我们计算模型预测与标注的真实值之间的匹配成本。

单次检测。给定输入图像和查询图像，模型需预测带有边界框的物体。这些物体应在语义上与查询图像相似。通常，查询图像由来自同一类别不同图像的补丁构建。在单次检测中，查询图像通过CNN（与输入图像共享）提取，并将补丁特征添加至所有物体查询中。在单次检测中，我们仅关注边界框的匹配结果（*i.e.* 是否与查询匹配），而非物体检测中的具体类别。因此，在损失和匹配成本计算中，单次检测采用二元分类输出，而非物体检测中的多类别分类。

4 实验

我们使用ImageNet [39]对模型进行预训练，并在PASCAL VOC [6]和COCO [5]上微调参数。在所有实验中，我们采用ResNet-50骨干网络、6层Transformer编码器、宽度为256且包含8个注意力头的6层解码器来实现UP-DETR模型（参数总量41.3M）。参考DETR²的开源代码，我们在提出的UP-DETR及我们重新实现的DETR中使用相同的超参数。我们将ResNet-50和ResNet-101简称为R50和R101。请注意，UP-DETR与DETR具有完全相同的模型架构、匹配代价和损失计算方式，因此两者的FLOPs、参数量及运行FPS完全一致。

预训练设置。UP-DETR在128万张ImageNet训练集上进行了无监督预训练，且未使用任何ImageNet标签。其CNN骨干网络（ResNet-50）同样采用SwAV策略[12]进行无监督预训练，且在Transformer预训练期间参数保持冻结。由于ImageNet输入图像尺寸较小，我们将其短边调整至[320, 480]像素范围内，同时长边不超过600像素。对于给定图像，我们以随机坐标、高度和宽度裁剪查询图像块，将其缩放至128×128像素，并采用SimCLR风格[14]进行变换（不含水平翻转），包括随机色彩失真与高斯模糊。在早期实验中，我们尝试了不同采样策略（*i.e.* 网格采样及不同超参数下的随机采样），但发现不同策略间未产生显著差异。因此，我们采用随机采样以验证方法的泛化能力。此外，使用AdamW[40]优化UP-DETR，初始学习率为 1×10^{-4} ，权重衰减为 1×10^{-4} 。我们在8块Nvidia V100 GPU上以256的批大小训练模型60个周期，并在第40周期将学习率乘以0.1。

微调设置。模型使用预训练的UP-DETR参数进行初始化，并在PASCAL VOC和COCO数据集上对所有参数（包括CNN）进行微调。我们对

表1：在PASCAL VOC trainval07+12上训练并在test2007上评估的目标检测结果。DETR和UP-DETR使用R50骨干网络，Faster R-CNN使用R50-C4骨干网络。括号中的数值表示与相同训练计划下DETR的差距。

Model/Epoch	AP	AP ₅₀	AP ₇₅
Faster R-CNN	56.1	82.6	62.7
DETR/150	49.9	74.5	53.1
UP-DETR/150	56.1 (+6.2)	79.7 (+5.2)	60.6 (+7.5)
DETR/300	54.1	78.0	58.3
UP-DETR/300	57.2 (+3.1)	80.1 (+2.1)	62.0 (+3.7)

模型初始学习率设置为：transformer部分为 1×10^{-4} ，CNN骨干网络部分为 5×10^{-5} ，其余设置与DETR[1]相同，使用八块V100 GPU，每块GPU处理四张图像。模型采用短/长周期调度进行微调，分别训练150/300个周期，并在第100/200个周期时将学习率乘以0.1。

预训练与微调之间的差距。物体级别的预训练可以提升物体检测任务中的微调性能[41]，但这类方法通常需要物体先验知识，*e.g.*，例如在预训练阶段使用选择性搜索算法获取物体候选框。在我们的工作中，我们在ImageNet数据集上以图像块级别进行预训练，因为这是在ImageNet上预训练并迁移到COCO或VOC数据集的常见做法[10][37]。然而，预训练与微调之间的差距始终存在。例如，ImageNet图像大多包含单个物体，而COCO图像通常包含多个物体；ImageNet图像尺寸相对较小，而COCO图像尺寸较大。此外，本工作中的预训练任务采用随机采样的图像块级别进行，但微调任务是在带标注数据上进行监督训练的。尽管预训练与微调之间的上述差距无法完全消除，但我们的方法通过随机采样策略展现出良好的泛化能力，且无需任何先验知识。

4.1 PASCAL VOC 目标检测

设置。该模型在PASCAL VOC trainval07+12（~16.5k张图像）上进行微调，并在test2007上评估。我们报告了COCO风格的指标，包括AP、AP₅₀（默认VOC指标）和AP₇₅。为了全面比较，我们还报告了使用R50-C4骨干网络[12]的Faster R-CNN的结果，其性能远优于R50（C5阶段）[42]。与R50相比，采用R50-C4的DETR显著增加了计算成本，因此我们使用R50骨干网络对UP-DETR进行微调。即便如此，UP-DETR仍表现良好。为了强调预训练模型的有效性，我们报告了DETR（从随机初始化的transformer开始）和UP-DETR（从预训练的transformer开始）在150和300轮训练后的结果。

结果。表1展示了在PASCAL VOC数据集上的目标检测结果。我们发现DETR在PASCAL VOC上表现不佳，各项指标均大幅落后于Faster R-CNN。由于VOC数据规模相对较小，UP-DETR的预训练Transformer显著提升了DETR在短期和长期训练计划中的性能：在150（300）轮训练中，AP分别提升至+6.2（+3.1）、+5.2（+2.1）₅₀和+7.5（+3.7）₇₅。此外，UP-DETR（R50）取得了与Faster R-CNN（R50-C4）相当的结果，且AP更优。

2. <https://github.com/facebookresearch/detr>

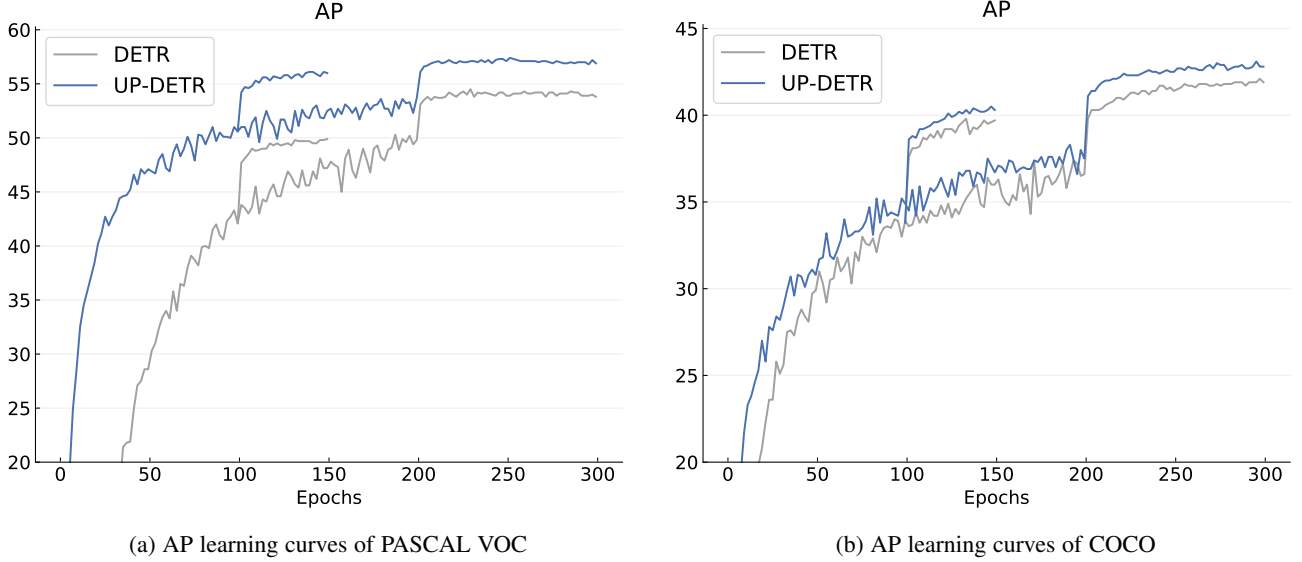


Fig. 4: AP (COCO style) learning curves of DETR and UP-DETR on PASCAL VOC and COCO datasets. Models are trained with the SwAV pre-training ResNet-50 for 150 and 300 epochs, and the learning rate is reduced at 100 and 200 epochs, respectively.

TABLE 2: Object detection results trained on COCO train2017 and evaluated on val2017. Faster R-CNN, DETR and UP-DETR are performed under comparable settings. † for values evaluated on COCO test-dev, which are always slightly higher than val2017. The values in the brackets are the gaps compared to DETR (SwAV CNN) with the same training schedule.

Model	Backbone	Epochs	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster R-CNN † [43]	R101-FPN	-	36.2	59.1	39.0	18.2	39.0	48.2
Mask R-CNN † [44]	R101-FPN	-	38.2	60.3	41.7	20.1	41.1	50.2
Grid R-CNN † [45]	R101-FPN	-	41.5	60.9	44.5	23.3	44.9	53.1
Double-head R-CNN [27]	R101-FPN	-	41.9	62.4	45.9	23.9	45.2	55.8
RetinaNet † [20]	R101-FPN	-	39.1	59.1	42.3	21.8	42.7	50.2
FCOS † [22]	R101-FPN	-	41.5	60.7	45.0	24.4	44.8	51.6
DETR [1]	R50	500	42.0	62.4	44.2	20.5	45.8	61.1
Faster R-CNN	R50-FPN	3×	40.2	61.0	43.8	24.2	43.5	52.0
DETR (Supervised CNN)	R50	150	39.5	60.3	41.4	17.5	43.0	59.1
DETR (SwAV CNN) [12]	R50	150	39.7	60.3	41.7	18.5	43.8	57.5
UP-DETR	R50	150	40.5 (+0.8)	60.8	42.6	19.0	44.4	60.0
Faster R-CNN	R50-FPN	9×	42.0	62.1	45.5	26.6	45.4	53.4
DETR (Supervised CNN)	R50	300	40.8	61.2	42.9	20.1	44.5	60.3
DETR (SwAV CNN) [12]	R50	300	42.1	63.1	44.5	19.7	46.3	60.9
UP-DETR	R50	300	43.1 (+1.0)	63.4	46.0	21.6	46.8	62.4

find that both UP-DETR and DETR perform a little worse than Faster R-CNN in AP₅₀ and AP₇₅. This may come from different ratios of feature maps (C4 for Faster R-CNN) and no NMS post-processing (NMS lowers AP but slightly improves AP₅₀).

Fig. 4a shows the AP (COCO style) learning curves on VOC. UP-DETR significantly speeds up the model convergence. After the learning rate reduced, UP-DETR significantly boosts the performance of DETR with a large AP improvement. Noting that UP-DETR obtains 56.1 AP after 150 epochs, however, its counterpart DETR (scratch transformer) only obtains 54.1 AP even after 300 epochs and does not catch up even training longer. It suggests that pre-training transformer is indispensable on insufficient training data (*i.e.* ~ 16.5 K images on VOC).

4.2 COCO Object Detection

Setup. The model is fine-tuned on COCO train2017 (~ 118 k images) and evaluated on val2017. There are lots of small objects in COCO dataset, where DETR performs poorly [1]. Therefore, we report AP, AP₅₀, AP₇₅, AP_S, AP_M and AP_L for a

comprehensive comparison. Moreover, we also report the results of highly optimized Faster R-CNN with feature pyramid network (FPN) with short (3 $\times$) and long (9 $\times$) training schedules, which are known to improve the performance results [46]. To avoid supervised CNN bringing supplementary information, we use SwAV pre-training CNN as the backbone of UP-DETR without any human annotations.

Results. Table 2 shows the object detection results on COCO dataset. With 150 epoch schedule, UP-DETR outperforms DETR (SwAV CNN) by 0.8 AP and achieves a comparable performance as compared with Faster R-CNN (R50-FPN) (3 $\times$ schedule). With 300 epoch schedule, UP-DETR obtains **43.1** AP on COCO, which is 1.0 AP better than DETR (SwAV CNN) and 1.1 AP better than Faster R-CNN (R50-FPN) (9 $\times$ schedule). Overall, UP-DETR comprehensively outperforms DETR in detection of small, medium and large objects with both short and long training schedules. Regrettably, UP-DETR is still slightly lagging behind Faster R-CNN in AP_S, because of the lacking of FPN-like architecture [43].

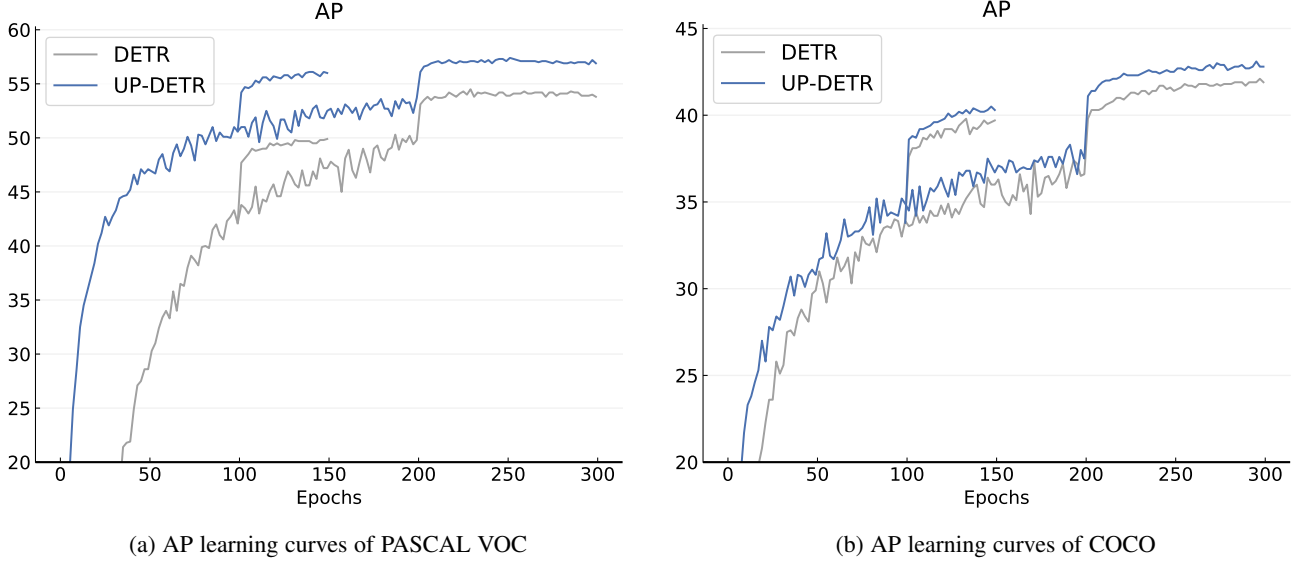


图 4 DETR与UP-DETR在PASCAL VOC和COCO数据集上的AP（COCO风格）学习曲线。模型采用SwAV预训练ResNet-50分别进行150和300个周期，学习率分别在100和200个周期时降低。

表2: 在COCO train2017上训练并在val2017上评估的目标检测结果。Faster R-CNN、DETR和UP-DETR均在可比较的设置下进行。 $\dagger$ 表示在COCO test-dev上评估的数值，这些数值通常略高于val2017的评估结果。括号中的数值表示与采用相同训练计划的DETR（SwAV CNN）相比的差距。

Model	Backbone	Epochs	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster R-CNN $\dagger$ [43]	R101-FPN	-	36.2	59.1	39.0	18.2	39.0	48.2
Mask R-CNN $\dagger$ [44]	R101-FPN	-	38.2	60.3	41.7	20.1	41.1	50.2
Grid R-CNN $\dagger$ [45]	R101-FPN	-	41.5	60.9	44.5	23.3	44.9	53.1
Double-head R-CNN [27]	R101-FPN	-	41.9	62.4	45.9	23.9	45.2	55.8
RetinaNet $\dagger$ [20]	R101-FPN	-	39.1	59.1	42.3	21.8	42.7	50.2
FCOS $\dagger$ [22]	R101-FPN	-	41.5	60.7	45.0	24.4	44.8	51.6
DETR [1]	R50	500	42.0	62.4	44.2	20.5	45.8	61.1
Faster R-CNN	R50-FPN	3 $\times$	40.2	61.0	43.8	24.2	43.5	52.0
DETR (Supervised CNN)	R50	150	39.5	60.3	41.4	17.5	43.0	59.1
DETR (SwAV CNN) [12]	R50	150	39.7	60.3	41.7	18.5	43.8	57.5
UP-DETR	R50	150	40.5 (+0.8)	60.8	42.6	19.0	44.4	60.0
Faster R-CNN	R50-FPN	9 $\times$	42.0	62.1	45.5	26.6	45.4	53.4
DETR (Supervised CNN)	R50	300	40.8	61.2	42.9	20.1	44.5	60.3
DETR (SwAV CNN) [12]	R50	300	42.1	63.1	44.5	19.7	46.3	60.9
UP-DETR	R50	300	43.1 (+1.0)	63.4	46.0	21.6	46.8	62.4

发现UP-DETR和DETR在AP₅₀和AP₇₅上的表现均略逊于Faster R-CNN。这可能源于特征图比例不同（Faster R-CNN使用C4特征图）以及缺少NMS后处理（NMS会降低AP但能轻微提升AP₅₀）。

图4a展示了在VOC数据集上的AP（COCO风格）学习曲线。UP-DETR显著加快了模型收敛速度。学习率下降后，UP-DETR大幅提升了DETR的性能，AP值有明显提高。值得注意的是，UP-DETR在150个训练周期后达到了56.1 AP，而其对照模型DETR（随机初始化的Transformer）即使在300个周期后也仅获得54.1 AP，且更长时间的训练仍未能追平。这表明在训练数据不足时（*i.e.* $\sim 16.5K$ 张VOC图像），对Transformer进行预训练是不可或缺的。

4.2 COCO 目标检测

设置。该模型在COCO train2017数据集（ $\sim 118k$ 张图像）上进行微调，并在val2017上进行评估。COCO数据集中存在大量小物体，而DETR在此类物体上表现不佳[1]。因此，我们报告了AP、AP₅₀、AP₇₅、AP_S、AP_M和AP_L的

全面比较。此外，我们还报告了采用特征金字塔网络（FPN）并经过短（3 $\times$ ）和长（9 $\times$ ）训练周期的高度优化Faster R-CNN的结果，这些方法已知能提升性能表现[46]。为避免监督式CNN引入额外信息，我们使用SwAV预训练的CNN作为UP-DETR的骨干网络，且未采用任何人工标注。

结果。表2展示了在COCO数据集上的目标检测结果。在150个训练周期的设定下，UP-DETR以0.8 AP的优势超越了DETR（SwAV CNN），并与Faster R-CNN（R50-FPN）（3 $\times$ 周期设定）取得了相当的性能。在300个训练周期的设定下，UP-DETR在COCO上获得了43.1 AP，这比DETR（SwAV CNN）高出1.0 AP，比Faster R-CNN（R50-FPN）（9 $\times$ 周期设定）高出1.1 AP。总体而言，无论在短期还是长期的训练设定下，UP-DETR在小、中、大目标的检测上都全面超越了DETR。遗憾的是，由于缺乏类似FPN的结构[43]，UP-DETR在AP_S上仍略微落后于Faster R-CNN。

TABLE 3: One-shot detection results trained on VOC 2007train val and 2012train val sets and evaluated on VOC 2007test set.

Model	seen class																unseen class					
	plant	sofa	tv	car	bottle	boat	chair	person	bus	train	horse	bike	dog	bird	mbike	table	AP⁵⁰	cow	sheep	cat	aero	AP⁵⁰
SiamFC [47]	3.2	22.8	5.0	16.7	0.5	8.1	1.2	4.2	22.2	22.6	35.4	14.2	25.8	11.7	19.7	27.8	15.1	6.8	2.28	31.6	12.4	13.3
SiamRPN [48]	1.9	15.7	4.5	12.8	1.0	1.1	6.1	8.7	7.9	6.9	17.4	17.8	20.5	7.2	18.5	5.1	9.6	15.9	15.7	21.7	3.5	14.2
CompNet [49]	28.4	41.5	65.0	66.4	37.1	49.8	16.2	31.7	69.7	73.1	75.6	71.6	61.4	52.3	63.4	39.8	52.7	75.3	60.0	47.9	25.3	52.1
CoAE [50]	30.0	54.9	64.1	66.7	40.1	54.1	14.7	60.9	77.5	78.3	77.9	73.2	80.5	70.8	72.4	46.2	60.1	83.9	67.1	75.6	46.2	68.2
Li <i>et al.</i> [51]	33.7	58.2	67.5	72.7	40.8	48.2	20.1	55.4	78.2	79.0	76.2	74.6	81.3	71.6	72.0	48.8	61.1	74.3	68.5	81.0	52.4	69.1
DETR	11.4	42.2	44.1	63.4	14.9	40.6	20.6	63.7	62.7	71.5	59.6	52.7	60.6	53.6	54.9	22.1	46.2	62.7	55.2	65.4	45.9	57.3
UP-DETR	46.7	61.2	75.7	81.5	54.8	57.0	44.5	80.7	74.5	86.8	79.1	80.3	80.6	72.0	70.9	57.8	69.0	80.9	71.0	80.4	59.9	73.1

TABLE 4: Panoptic segmentation results on the COCO val2017 dataset with the same ResNet-50 backbone. The PanopticFPN++, UPSNet and DETR results are re-implemented by Carion *et al.* [1].

Model	PQ	SQ	RQ	PQ th	SQ th	RQ th	PQ st	SQ st	RQ st	AP ^{seg}
PanopticFPN++ [52]	42.4	79.3	51.6	49.2	82.4	58.8	32.3	74.8	40.6	37.7
UPSNet [53]	42.5	78.0	52.5	48.6	79.4	59.6	33.4	75.9	41.7	34.3
UPSNet-M [53]	43.0	79.1	52.8	48.9	79.7	59.7	34.1	78.2	42.3	34.3
DETR [1]	44.3	80.0	54.5	49.2	80.6	60.3	37.0	79.1	45.9	32.9
UP-DETR	44.7	80.5	54.9	49.7	80.9	60.8	37.2	79.3	46.2	34.3

Fig. 4b shows the AP learning curves on COCO. UP-DETR outperforms DETR for both 150 and 300 epoch schedules with faster convergence. The performance improvement is more noticeable before reducing the learning rate. After reducing the learning rate, UP-DETR still holds the lead of DETR by ~ 1.0 AP improvement. It suggests that the pre-training transformer is still indispensable even on sufficient training data (*i.e.* $\sim 118K$ images on COCO).

4.3 One-Shot Detection

Setup. Given a query image patch whose class label is not included in the training data, one-shot detection aims to detect all instances with the same class in a target image. One-shot detection is a promising research direction that can detect unseen instances. With feeding query patches to the decoder, UP-DETR is naturally compatible to one-shot detection task. Therefore, one-shot detection can also be treated as a downstream fine-tuning task of UP-DETR.

Following the same one-shot detection setting as [50], we crop out the ground truth bounding boxes as the query image patches. During fine-tuning, for the given image, we randomly sample a query image patch of a seen class that exists in this image, and the training ground truth is filtered according to the query image patch. During evaluation, we first randomly shuffle the query image patches of the class (existing in the image) with a specific random seed of target image ID, then pick up the first five query image patches, and finally average their AP scores. Every existing class in each image is evaluated and then averaged. The shuffle procedure ensures the results in stable statistics for evaluation. We train/fine-tune DETR/UP-DETR on VOC 2007train val and 2012train val sets with 300 epochs, and evaluate on VOC 2007test set. We follow the same setting as Li *et al.* [51].

Results. Table 3 shows the comparison to the state-of-the-art one-shot detection methods. UP-DETR significantly boosts the performance of DETR on both seen (**+22.8 AP⁵⁰** gain) and unseen (**+15.8 AP⁵⁰** gain) classes. Moreover, UP-DETR outperforms all compared methods in both seen (min **+7.9 AP⁵⁰** gain) and unseen (min **+4.0 AP⁵⁰** gain) classes of one-shot detection. It further verifies the effectiveness of our pre-training pretext task.

TABLE 5: The ablation results of pre-training models with single-query patch and multi-query patches on PASCAL VOC. The values in the brackets are the gaps compared to DETR with the same training schedule.

Model	AP	AP ₅₀	AP ₇₅
DETR	49.9	74.5	53.1
UP-DETR (M=1)	53.1 (+3.2)	77.2 (+2.7)	57.4
UP-DETR (M=10)	54.9 (+5.0)	78.7 (+4.2)	59.1

4.4 Panoptic Segmentation

The original DETR can be easily extended to panoptic segmentation [52] by adding a mask head on top of the decoder outputs. Following the same panoptic segmentation training schema as DETR [1], we fine-tune UP-DETR with COCO panoptic segmentation annotation (extra stuff annotation) for only box annotations with 300 epochs. Then, we freeze all the weights of DETR and train the mask head for 25 epochs. We find that UP-DETR also boosts the fine-tuning performance of panoptic segmentation.

Table 4 shows the comparison to state-of-the-art methods on panoptic segmentation with the ResNet-50 backbone. As seen, UP-DETR outperforms DETR³ with **+0.4 PQ**, **+0.5 PQth** and **+1.4 AP^{seg}**.

4.5 Ablations

In ablation experiments, we train/pre-train DETR/UP-DETR models for 15 epochs with the learning rate multiplied by 0.1 at the 10th epoch. We fine-tune the UP-DETR models on PASCAL VOC following the setup in Section 4.1 with 150 epochs. Therefore, the average precision results in ablations are relatively lower than those shown in Section 4.1.

4.5.1 Single-query patch vs. Multi-query patches

We pre-train the UP-DETR model with single-query patch ($M = 1$) and multi-query patches ($M = 10$). Other hyper-parameters are set as mentioned above.

3. With a bug fixed in github.com/facebookresearch/detr/issues/247, the DETR baseline is better than paper report.

表3: 在VOC 2007训练验证集和2012训练验证集上训练, 并在VOC 2007测试集上评估的单次检测结果。

Model	seen class																	unseen class				
	plant	sofa	tv	car	bottle	boat	chair	person	bus	train	horse	bike	dog	bird	mbike	table	AP ⁵⁰	cow	sheep	cat	aero	AP ⁵⁰
SiamFC [47]	3.2	22.8	5.0	16.7	0.5	8.1	1.2	4.2	22.2	22.6	35.4	14.2	25.8	11.7	19.7	27.8	15.1	6.8	2.28	31.6	12.4	13.3
SiamRPN [48]	1.9	15.7	4.5	12.8	1.0	1.1	6.1	8.7	7.9	6.9	17.4	17.8	20.5	7.2	18.5	5.1	9.6	15.9	15.7	21.7	3.5	14.2
CompNet [49]	28.4	41.5	65.0	66.4	37.1	49.8	16.2	31.7	69.7	73.1	75.6	71.6	61.4	52.3	63.4	39.8	52.7	75.3	60.0	47.9	25.3	52.1
CoAE [50]	30.0	54.9	64.1	66.7	40.1	54.1	14.7	60.9	77.5	78.3	77.9	73.2	80.5	70.8	72.4	46.2	60.1	83.9	67.1	75.6	46.2	68.2
Li et al. [51]	33.7	58.2	67.5	72.7	40.8	48.2	20.1	55.4	78.2	79.0	76.2	74.6	81.3	71.6	72.0	48.8	61.1	74.3	68.5	81.0	52.4	69.1
DETR	11.4	42.2	44.1	63.4	14.9	40.6	20.6	63.7	62.7	71.5	59.6	52.7	60.6	53.6	54.9	22.1	46.2	62.7	55.2	65.4	45.9	57.3
UP-DETR	46.7	61.2	75.7	81.5	54.8	57.0	44.5	80.7	74.5	86.8	79.1	80.3	80.6	72.0	70.9	57.8	69.0	80.9	71.0	80.4	59.9	73.1

表4: 在COCO val2017数据集上使用相同ResNet-50骨干网络的全景分割结果。PanopticFPN++、UPNet和DETR的结果由Carion *et al*重新实现。[1]。

Model	PQ	SQ	RQ	PQ th	SQ th	RQ th	PQ st	SQ st	RQ st	AP ^{seg}
PanopticFPN++ [52]	42.4	79.3	51.6	49.2	82.4	58.8	32.3	74.8	40.6	37.7
UPNet [53]	42.5	78.0	52.5	48.6	79.4	59.6	33.4	75.9	41.7	34.3
UPNet-M [53]	43.0	79.1	52.8	48.9	79.7	59.7	34.1	78.2	42.3	34.3
DETR [1]	44.3	80.0	54.5	49.2	80.6	60.3	37.0	79.1	45.9	32.9
UP-DETR	44.7	80.5	54.9	49.7	80.9	60.8	37.2	79.3	46.2	34.3

图4b展示了在COCO数据集上的平均精度学习曲线。UP-DETR在150和300轮训练计划中均优于DETR, 且收敛速度更快。在降低学习率之前, 性能提升更为明显。降低学习率后, UP-DETR仍以~1.0 AP的优势领先于DETR。这表明即使在充足训练数据(COCO数据集包含*i.e.* ~ 118K图像)的情况下, 预训练Transformer仍然不可或缺。

4.3 单次检测

设置。给定一个查询图像块, 其类别标签未包含在训练数据中, 单次检测的目标是在目标图像中检测出所有属于同一类别的实例。单次检测是一个有前景的研究方向, 能够检测未见过的实例。通过将查询块输入解码器, UP-DETR天然适用于单次检测任务。因此, 单次检测也可视为UP-DETR的下游微调任务。

遵循与[50]相同的单次检测设置, 我们将真实边界框裁剪为查询图像块。在微调过程中, 对于给定图像, 我们随机采样该图像中存在的已见类别的查询图像块, 并根据查询图像块筛选训练真实标签。评估时, 我们首先使用目标图像ID的特定随机种子对图像中存在的该类别的查询图像块进行随机打乱, 然后选取前五个查询图像块, 最后计算其平均精度(AP)得分的平均值。对每张图像中存在的每个类别进行评估后取平均。打乱过程确保了评估统计结果的稳定性。我们在VOC 2007train val和2012train val数据集上以300个周期训练/微调DETR/UP-DETR, 并在VOC 2007test集上评估。我们遵循与Li *et al*相同的设置[51]。

结果。表3展示了与最先进的一次性检测方法的比较。UP-DETR显著提升了DETR在已见类别(+22.8 AP⁵⁰提升)和未见类别(+15.8 AP⁵⁰提升)上的性能。此外, 在一次性检测的已见类别(最小+7.9 AP⁵⁰提升)和未见类别(最小+4.0 AP⁵⁰提升)中, UP-DETR均优于所有对比方法。这进一步验证了我们预训练前置任务的有效性。

表5: 在PASCAL VOC上使用单查询补丁与多查询补丁的预训练模型消融结果。括号中的数值是与相同训练计划下DETR相比的差距。

Model	AP	AP ₅₀	AP ₇₅
DETR	49.9	74.5	53.1
UP-DETR (M=1)	53.1 (+3.2)	77.2 (+2.7)	57.4
UP-DETR (M=10)	54.9 (+5.0)	78.7 (+4.2)	59.1

4.4 全景分割

原始的DETR可以通过在解码器输出之上添加一个掩码头, 轻松扩展到全景分割[52]。遵循与DETR[1]相同的全景分割训练方案, 我们使用COCO全景分割标注(额外的物品标注)对UP-DETR进行微调, 仅使用边界框标注训练300个周期。随后, 我们冻结DETR的所有权重, 并训练掩码头25个周期。我们发现, UP-DETR也提升了全景分割的微调性能。

表4展示了使用ResNet-50骨干网络在全景分割任务上与前沿方法的对比。可以看出, UP-DETR以+0.4 PQ、+0.5 PQth和+1.4 AP^{seg}的优势超越了DETR³。

4.5 消融实验

在消融实验中, 我们以15个周期训练/预训练DETR/UP-DETR模型, 并在第10th个周期将学习率乘以0.1。我们按照第4.1节中的设置, 在PASCAL VOC数据集上对UP-DETR模型进行了150个周期的微调。因此, 消融实验中的平均精度结果相对较低于第4.1节中展示的结果。

4.5.1 Single-query patch vs. Multi-query patches

我们使用单查询补丁($M = 1$)和多查询补丁($M = 10$)对UP-DETR模型进行预训练。其他超参数设置如上所述。

3. 随着github.com/facebookresearch/detr/issues/247中的一个错误被修复, DETR基线表现优于论文报告。

TABLE 6: Ablation study on the frozen CNN and feature reconstruction for pre-training models with AP₅₀. The experiments are fine-tuned on PASCAL VOC with 150 epochs.

Case	Frozen CNN	Feature Reconstruction	AP ₅₀
DETR		scratch transformer	74.5
(a)			74.0
(b)	✓		78.7
(c)		✓	62.0
(d) UP-DETR	✓	✓	78.7

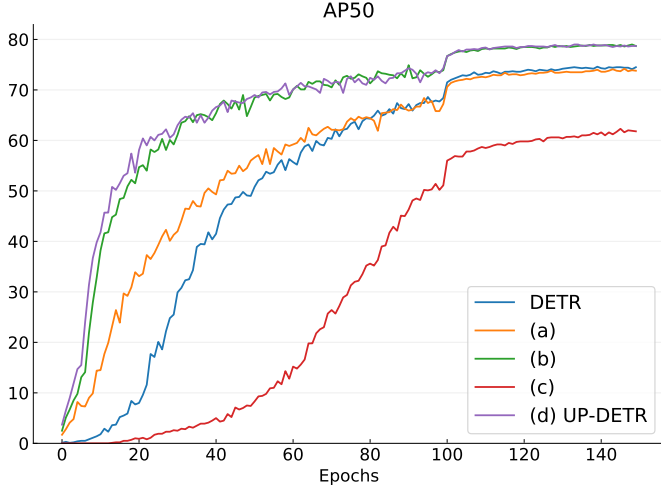


Fig. 5: Learning curves (AP₅₀) of DETR and four different pre-training models on PASCAL VOC trained with 150 epochs. The models correspond to the models in Table 6 one-to-one.

Table 5 shows the results of single-query patch and multi-query patches. Compared with DETR, UP-DETR surpasses it in all AP metrics by a large margin no matter with single-query patch or multi-query patches. When pre-training UP-DETR with the different number of query patches, UP-DETR ($M = 10$) performs better than UP-DETR ($M = 1$) on the fine-tuning task, although there are about 2.3 instances per image on PASCAL VOC. Therefore, we adopt the same UP-DETR with $M = 10$ for both PASCAL VOC and COCO instead of varying M for different downstream tasks.

4.5.2 Frozen CNN and Feature Reconstruction

To illustrate the importance of frozen CNN backbone and patch feature reconstruction of UP-DETR, we pre-train four different models with different combinations of whether freezing CNN and whether adding feature reconstruction. Note that all the models (including DETR) use the pre-trained CNN on ImageNet.

Table 6 shows AP₅₀ of DETR and four different pre-training models on PASCAL VOC with 150 epochs. As shown in Table 6, not all pre-trained models are better than DETR, but pre-training models (b) and (d) perform better than the others. More importantly, without frozen CNN, pre-training models (a) and (c) even perform worse than DETR. This confirms that the frozen pre-training backbone is essential to pre-train the transformer. In addition, it further confirms that the pretext task (random query patch detection) may weaken the feature discrimination of the pre-training CNN, and localization and classification have different feature preferences [26], [27], [28].

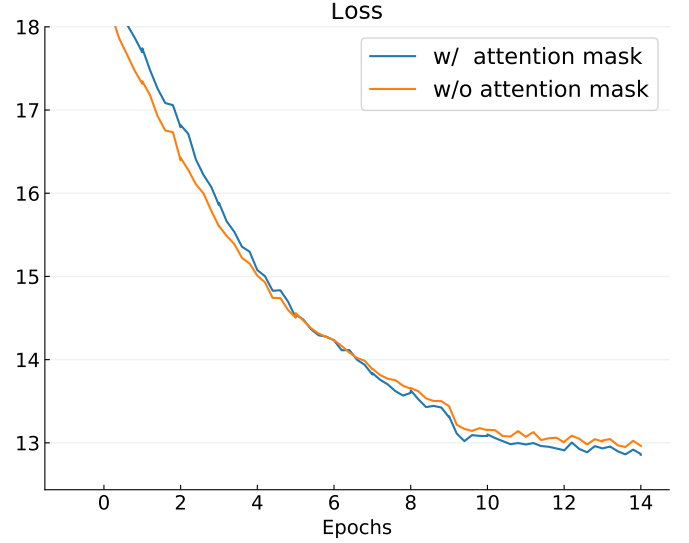


Fig. 6: The loss curves of pre-training procedure for UP-DETR w/ and w/o the attention mask.

Fig. 5 plots the AP₅₀ learning curves of DETR and four different pre-training models, where the models in Fig. 5 correspond to the models in Table 6 one-to-one. As shown in Fig. 5, (d) UP-DETR model achieves faster convergence speed at the early training stage with feature reconstruction. The experiment results suggest that random query patch detection is complementary to the contrastive learning for a better visual representation. The former is designed for the spatial localization with position embeddings, and the latter is designed for instance or cluster classification.

It is worth noting that UP-DETR with frozen CNN and feature reconstruction heavily relies on a pre-trained CNN model, *e.g.* SwAV CNN. Therefore, we believe that it is a promising direction to further investigate UP-DETR with random query patch detection and contrastive learning together to pre-train the whole DETR model from scratch.

4.5.3 Attention Mask

After downstream task fine-tuning, we find that there is no noticeable difference between the UP-DETR models pre-trained with and without attention mask. Instead of the fine-tuning results, we plot the loss curves in the pretext task to illustrate the effectiveness of attention mask.

As shown in Fig. 6, at the early training stage, UP-DETR without attention mask has a lower loss. However, as the model converging, UP-DETR with attention mask overtakes it with a lower loss. The curves seem weird at the first glance, but it is reasonable because the loss is calculated by the optimal bipartite matching. During the early training stage, the model is not converged, and the model without attention mask takes more object queries into attention. Intuitively, the model is easier to be optimized due to introducing more object queries. However, there is a mismatching between the query patch and the ground truth for the model without attention mask. As the model converging, the attention mask gradually takes effect, which masks the unrelated query patches and leads to a lower loss.

表6: 使用 AP_{50} 对预训练模型进行冻结CNN和特征重建的消融研究。实验在PASCAL VOC数据集上进行了150个epoch的微调。

Case	Frozen CNN	Feature Reconstruction	AP_{50}
DETR		scratch transformer	74.5
(a)			74.0
(b)	✓		78.7
(c)		✓	62.0
(d) UP-DETR	✓	✓	78.7

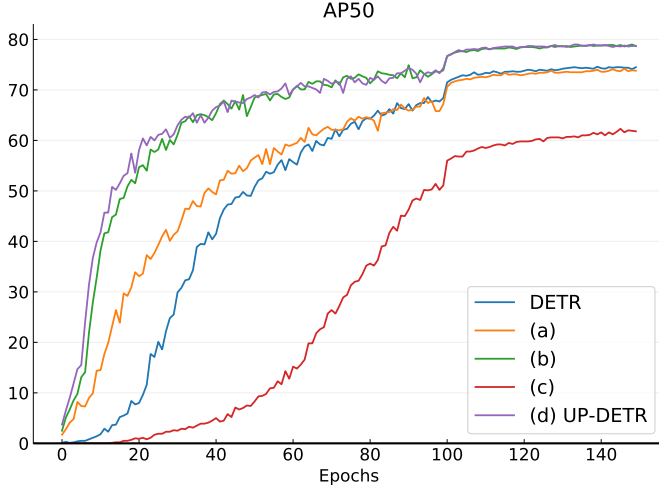


图5: DETR与四种不同预训练模型在PASCAL VOC上训练150个epoch的学习曲线 (AP_{50})。这些模型与表6中的模型一一对应。

表5展示了单查询补丁和多查询补丁的结果。与DETR相比, 无论使用单查询补丁还是多查询补丁, UP-DETR在所有AP指标上都大幅超越前者。当使用不同数量的查询补丁对UP-DETR进行预训练时, 尽管PASCAL VOC数据集中每张图像平均仅有约2.3个实例, UP-DETR ($\{v^*\} 10$)在微调任务上的表现仍优于UP-DETR ($\{v^*\} 1$)。因此, 我们在PASCAL VOC和COCO数据集上均采用相同的UP-DETR ($\{v^*\} 10$)配置, 而非针对不同下游任务调整 $\{v^*\}$ 数量。

4.5.2 Frozen CNN and Feature Reconstruction

为了说明冻结CNN主干网络和UP-DETR中图像块特征重建的重要性, 我们预训练了四种不同模型, 这些模型在是否冻结CNN以及是否添加特征重建方面采用了不同组合。请注意, 所有模型 (包括DETR) 都使用了在ImageNet上预训练的CNN。

表6展示了DETR及四种不同预训练模型在PASCAL VOC数据集上训练150轮后的 AP_{v^*} 性能。如表6所示, 并非所有预训练模型都优于DETR, 但预训练模型(b)和(d)的表现优于其他模型。更重要的是, 当CNN未冻结时, 预训练模型(a)和(c)的表现甚至比DETR更差。这证实了冻结预训练骨干网络对于预训练Transformer至关重要。此外, 该结果进一步表明前置任务 (随机查询补丁检测) 可能削弱预训练CNN的特征判别能力, 且定位与分类任务具有不同的特征偏好[26][27][28]。

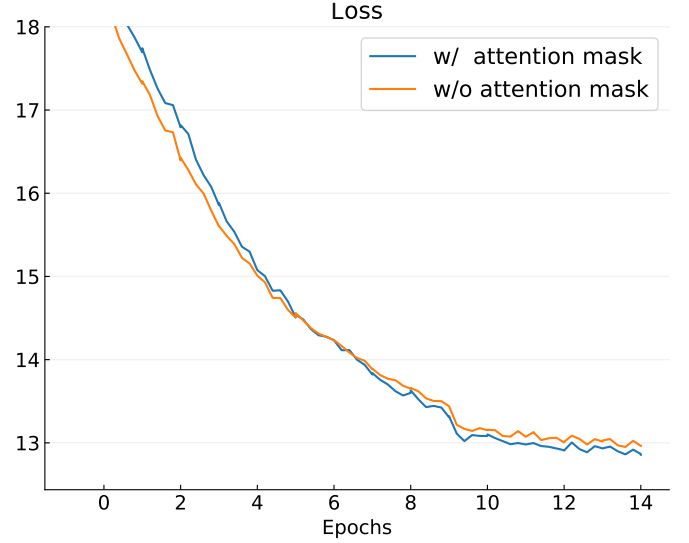


图6: 带与不带注意力掩码的UP-DETR预训练过程的损失曲线。

图5绘制了DETR及四种不同预训练模型的 AP_{50} 学习曲线, 其中图5中的模型与表6中的模型一一对应。如图5所示, (d) UP-DETR模型在早期训练阶段通过特征重建实现了更快的收敛速度。实验结果表明, 随机查询补丁检测与对比学习相辅相成, 有助于获得更好的视觉表示。前者通过位置嵌入设计用于空间定位, 后者则设计用于实例或聚类分类。

值得注意的是, 采用冻结CNN和特征重建的UP-DETR严重依赖于预训练的CNN模型e.g. SwAV CNN。因此, 我们认为将随机查询补丁检测与对比学习相结合, 进一步研究UP-DETR以从头预训练整个DETR模型, 是一个具有前景的方向。

4.5.3 Attention Mask

在下游任务微调后, 我们发现使用注意力掩码与不使用注意力掩码预训练的UP-DETR模型之间没有显著差异。我们并未展示微调结果, 而是绘制了预训练任务中的损失曲线, 以说明注意力掩码的有效性。

如图6所示, 在训练初期, 未使用注意力掩码的UP-DETR损失值较低。但随着模型逐渐收敛, 使用注意力掩码的UP-DETR反而以更低损失值实现反超。这些曲线初看有些反常, 但其实是合理的, 因为损失值是通过最优二分匹配计算得出的。在训练早期阶段, 模型尚未收敛, 未使用注意力掩码的模型会让更多物体查询参与注意力计算。直观来看, 引入更多物体查询会使模型更容易优化。然而, 对于未使用注意力掩码的模型, 查询图像块与真实标注之间会存在匹配偏差。随着模型不断收敛, 注意力掩码逐渐发挥作用, 通过屏蔽不相关的查询图像块, 最终实现了更低的损失值。

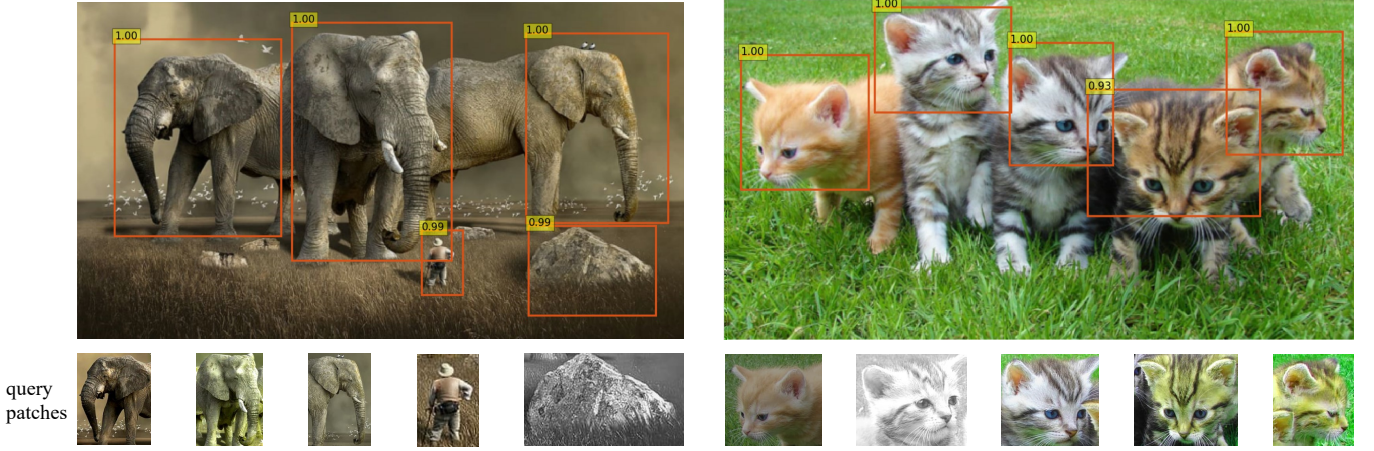


Fig. 7: The unsupervised localization of patch queries with UP-DETR. The first line is the original image with predicted bounding boxes. The second line is query patches cropped from the original image with data augmentation (color jitter and resizing). The value in the upper left corner of the bounding box is the model confidence. As seen, without any annotations, UP-DETR learns to detect patches with given queries in the unsupervised way.

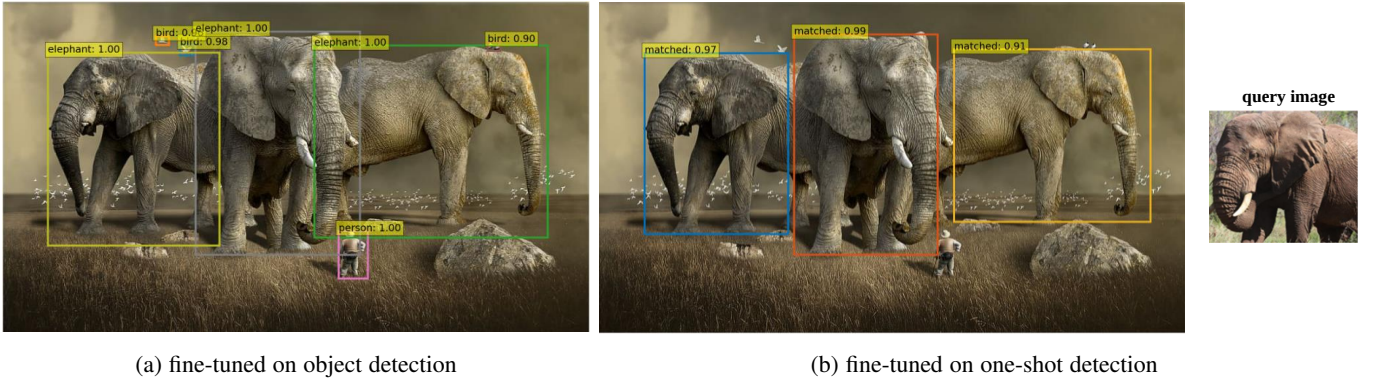


Fig. 8: The visualization result of fine-tuned experiments on object detection and one-shot detection. These two models are fine-tuned with the same UP-DETR model. (a) With the input image, the object detection model detects the objects with bounding boxes, whose classes exist in the training set. (b) Given input image and query image, the one-shot detection model detects the objects, which are semantically similar to the query image (with the matched tag instead of the class name).

4.6 Visualization

4.6.1 Pre-training

To further illustrate the ability of the pre-training model, we visualize the unsupervised localization results of given patch queries. Specifically, for the given image, we manually crop several object patches and apply the data augmentation to them. Then, we feed these patches as queries to the model. Finally, we visualize the model output with bounding boxes, whose classification confidence is greater than 0.9. This procedure can be treated as *unsupervised one-shot detection* or deep learning based *template matching*.

As shown in Fig. 7, pre-trained with random query patch detection, UP-DETR successfully learns to locate the bounding box of given query patches and suppresses the duplicated bounding boxes⁴. It demonstrates that UP-DETR with random query patch detection is effective to learn the ability of object localization.

4. Base picture credit: <https://www.piqsels.com/en/public-domain-photo-jrkkq>, <https://www.piqsels.com/en/public-domain-photo-smdfn>.

4.6.2 Fine-tuning

Fig. 8 shows the visualization result of fine-tuned models on object detection and one-shot detection. Our fine-tuning models perform well on these two tasks. For object detection task, the model is fine-tuned to detect objects with bounding boxes and classes. These classes (e.g. elephant, bird, person in Fig. 8a) must exist in the training set. Differently, for one-shot detection, given the input image and the query image, the model only predicts the similar objects with bounding boxes and 0/1 labels (matching or not). These two tasks can be unified in our UP-DETR framework by changing the input of the transformer decoder.

5 CONCLUSION

We present a novel pretext task called random query patch detection to unsupervisedly pre-train the transformer in DETR. With unsupervised pre-training, UP-DETR significantly outperforms DETR by a large margin with higher precision and much faster convergence on PASCAL VOC. For the challenging COCO dataset with sufficient training data, UP-DETR still surpasses DETR even with a long training schedule. It indicates that the

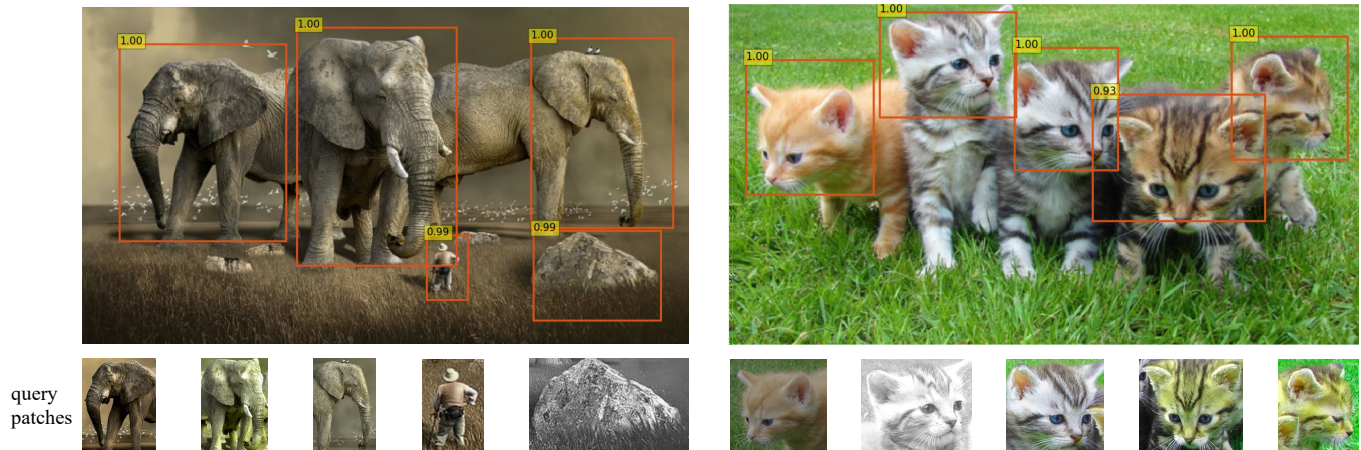


图7: 使用UP-DETR对图像块查询进行无监督定位。第一行为原始图像及预测边界框, 第二行为经过数据增强(色彩抖动与尺寸调整)后从原图裁剪的查询图像块。边界框左上角的数值为模型置信度。如图所示, 在没有任何标注的情况下, UP-DETR能以无监督方式学习检测给定查询对应的图像块。

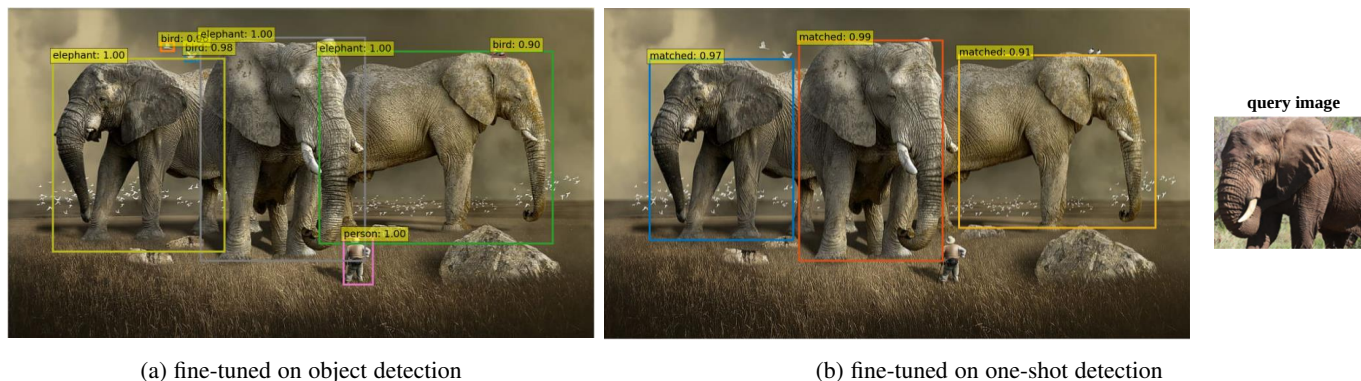


图8: 目标检测与单样本检测微调实验的可视化结果。这两个模型均使用相同的UP-DETR模型进行微调。(a) 对于输入图像, 目标检测模型通过边界框检测训练集中存在的类别物体。(b) 给定输入图像和查询图像, 单样本检测模型检测与查询图像语义相似的物体(使用匹配标签而非类别名称)。

4.6 可视化

4.6.1 Pre-training

为了进一步说明预训练模型的能力, 我们可视化给定图像块查询的无监督定位结果。具体而言, 对于给定图像, 我们手动裁剪出若干物体图像块, 并对它们进行数据增强处理。随后, 我们将这些图像块作为查询输入模型。最后, 我们将模型输出以边界框形式可视化, 这些边界框的分类置信度均高于0.9。此过程可视为 *unsupervised one-shot detection* 或基于深度学习的 *template matching*。

如图7所示, 通过随机查询补丁检测进行预训练, UP-DETR成功学会了定位给定查询补丁的边界框, 并抑制了重复的边界框⁴。这表明, 采用随机查询补丁检测的UP-DETR能有效学习目标定位能力。

4. Base picture credit: <https://www.piqsels.com/en/public-domain-photo-jrkkq>, <https://www.piqsels.com/en/public-domain-photo-smdfn>.

4.6.2 Fine-tuning

图8展示了微调模型在目标检测和单样本检测上的可视化结果。我们的微调模型在这两项任务中表现良好。对于目标检测任务, 模型经过微调后能够通过边界框和类别检测物体。这些类别(e.g. 图8a中的大象、鸟类、人物)必须存在于训练集中。不同的是, 对于单样本检测, 给定输入图像和查询图像, 模型仅通过边界框和0/1标签(匹配与否)预测相似物体。通过改变Transformer解码器的输入, 这两项任务可以在我们的UP-DETR框架中实现统一。

5 结论

我们提出了一种名为随机查询补丁检测的新颖前置任务, 用于对DETR中的Transformer进行无监督预训练。通过无监督预训练, UP-DETR在PASCAL VOC数据集上以更高的精度和更快的收敛速度显著超越DETR。对于具有充足训练数据的挑战性COCO数据集, 即使经过长时间的训练调度, UP-DETR仍然优于DETR。这表明

pre-training transformer is indispensable for different scales of training data in object detection. Besides, UP-DETR also provides a unified perspective for one-shot detection. It significantly boosts the performance on one-shot detection task.

From the perspective of unsupervised pre-training models, the pre-training CNN backbone and the pre-training transformer are separated now. Recent studies of unsupervised pre-training mainly focus on feature discrimination with contrastive learning instead of specialized modules for spatial localization. But in UP-DETR pre-training, the pretext task is mainly designed for patch localization by positional encodings and learnable object queries. We hope an advanced method can integrate the CNN and transformer pre-training into a unified end-to-end framework and apply UP-DETR to more downstream tasks (e.g. few-shot object detection and object tracking).

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预训练Transformer对于不同规模的目标检测训练数据都是不可或缺的。此外，UP-DETR还为单样本检测提供了统一视角，显著提升了单样本检测任务的性能。

从无监督预训练模型的角度来看，预训练的CNN主干网络和预训练的Transformer目前是分离的。近期的无监督预训练研究主要集中于通过对比学习进行特征判别，而非针对空间定位的专用模块。但在UP-DETR的预训练中，前置任务主要通过位置编码和可学习的对象查询来设计，以实现图像块定位。我们希望有一种先进方法能够将CNN和Transformer的预训练整合到一个统一的端到端框架中，并将UP-DETR应用于更多下游任务（e.g. 少样本目标检测与目标跟踪）。

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